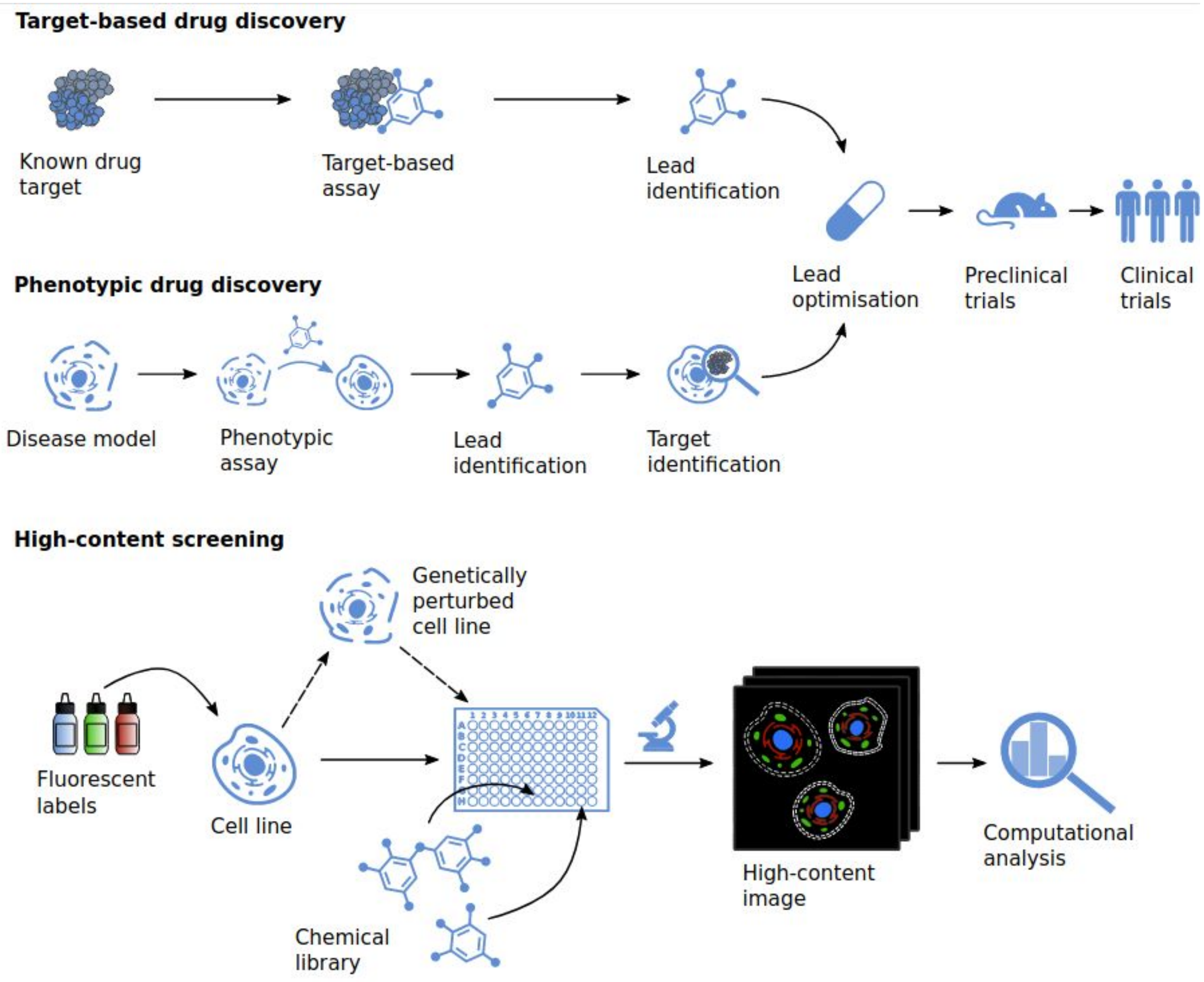
What can we do if there are no good targets

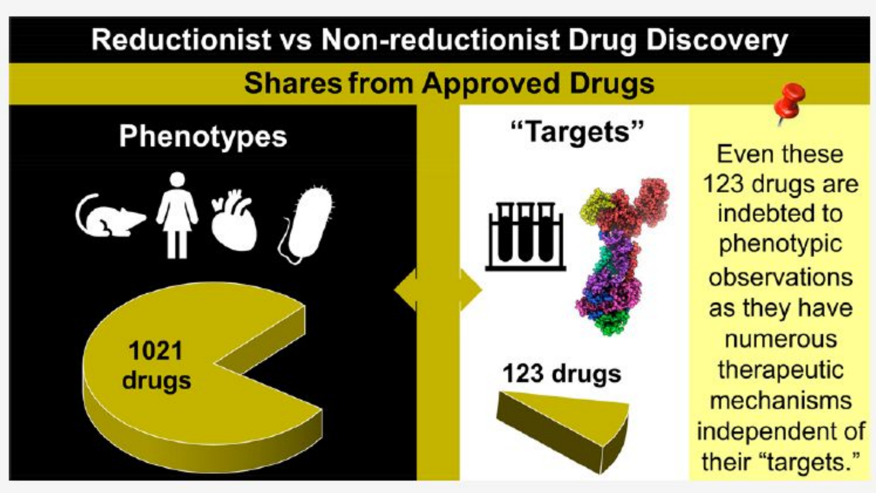
Mathematical and Computational Biology in Drug Discovery Module II

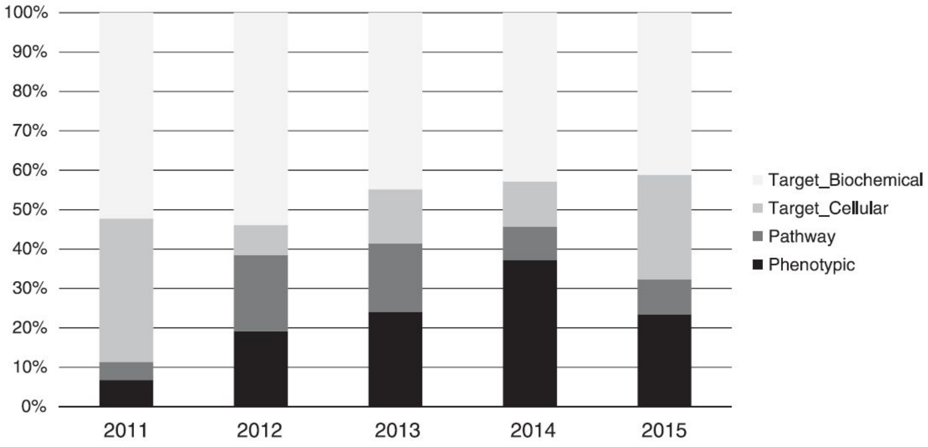
Dr. Jitao David Zhang
March-April 2026



Krentzel, Daniel, Spencer L. Shorte, and Christophe Zimmer. “Deep Learning in Image-Based Phenotypic Drug Discovery.” Trends in Cell Biology 33, no. 7 (2023): 538–54. https://doi.org/10.1016/j.tcb.2022.11.011.

Is target-based drug discovery the only way?



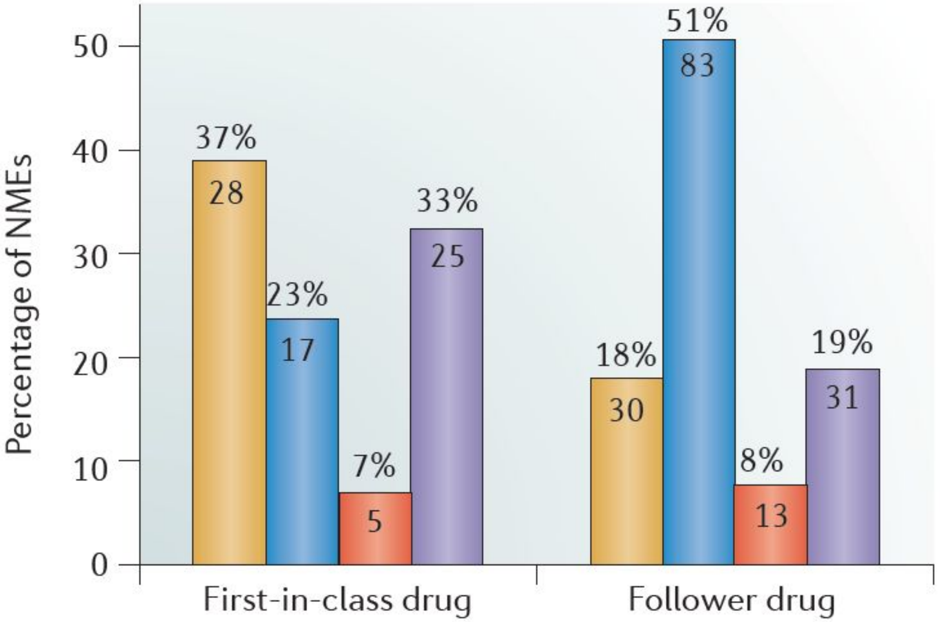


Sadri, Arash. 2023. "Is Target-Based Drug Discovery Efficient? Discovery and 'Off-Target' Mechanisms of All Drugs." Journal of Medicinal Chemistry 66 (18): 12651–77.

Haasen, Dorothea, Ulrich Schopfer, Christophe Antczak, Chantale Guy, Florian Fuchs, and Paul Selzer. 2017. "How Phenotypic Screening Influenced Drug Discovery: Lessons from Five Years of Practice." ASSAY and Drug Development Technologies 15 (6): 239–46.

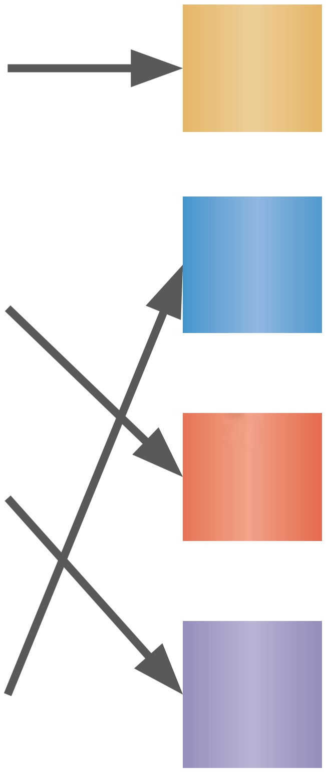
Five strategies when no good target is found

1. Phenotypic drug discovery
2. Natural products
3. Biologics
4. Interaction-based (multispecific) drug discovery
5. Drug repurposing or combination studies

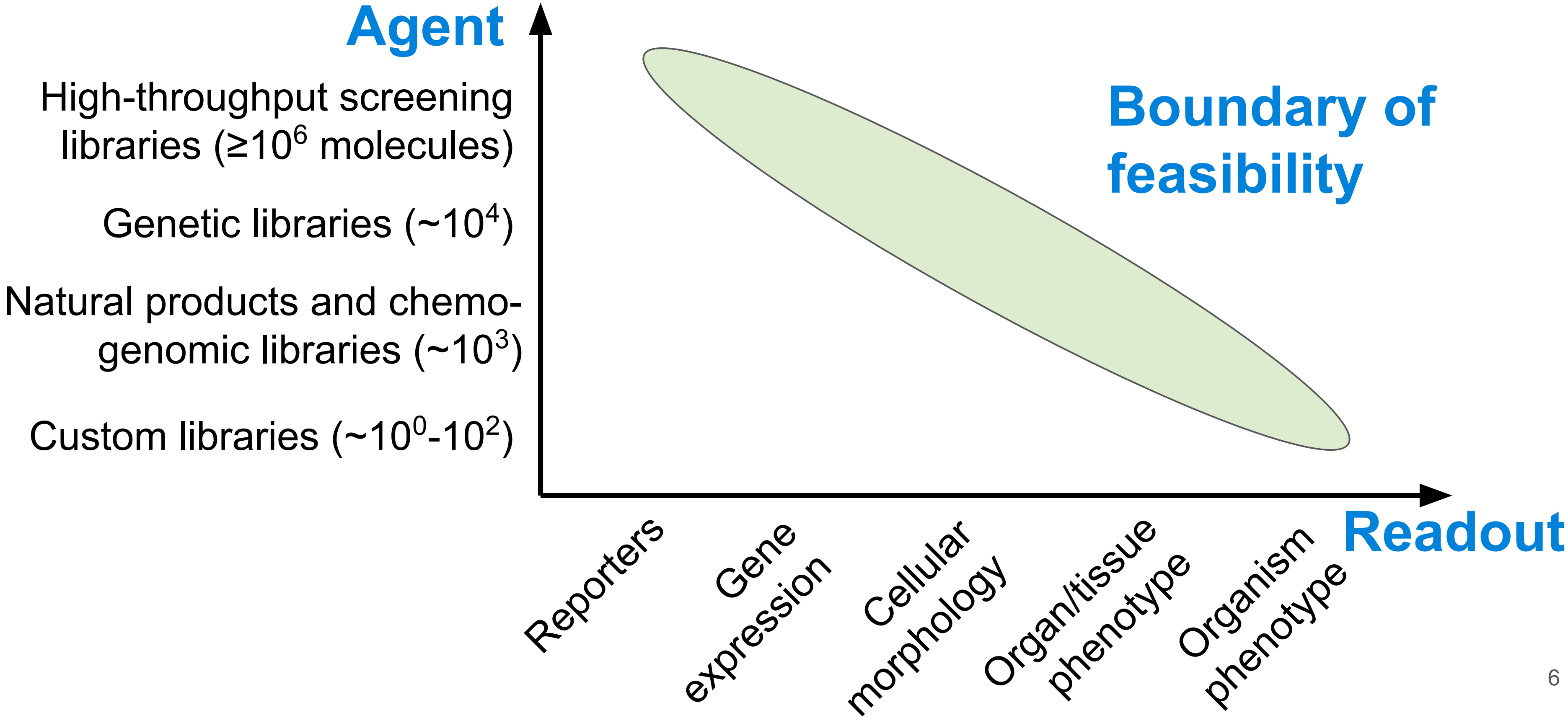


Connect the lines!

- Phenotypic screening
- Modified natural products
- Biologics
- Target-based screening



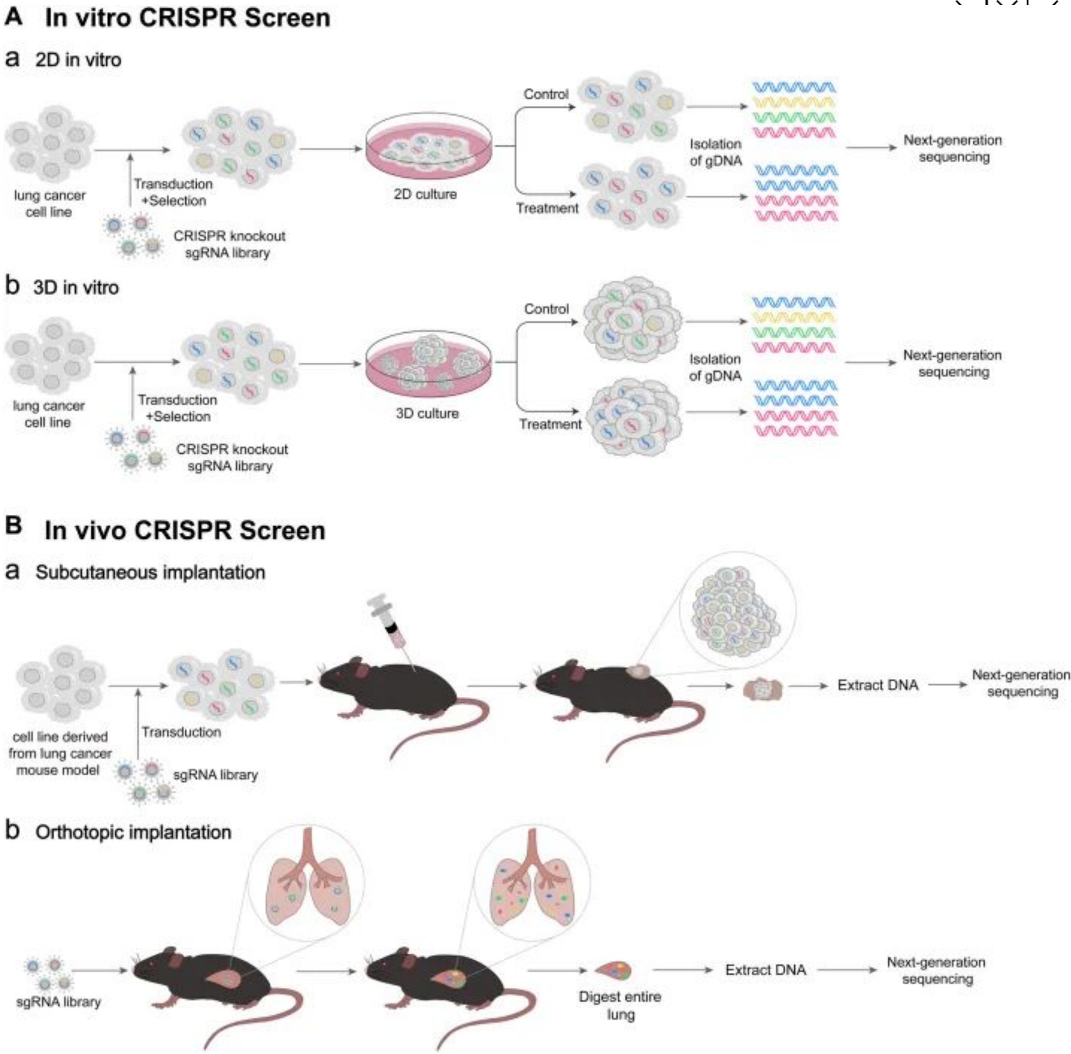
Phenotypic screenings by agent and readout



Genetic screening

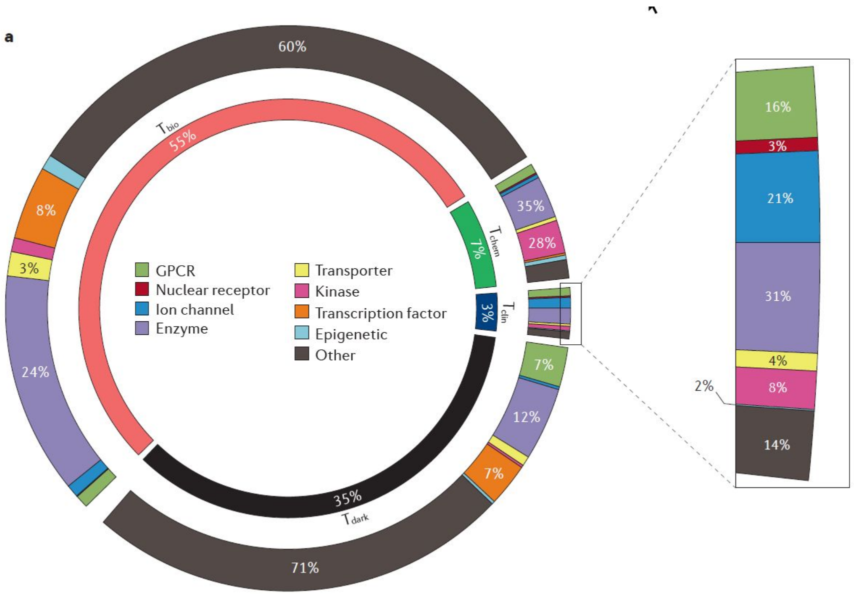
CRISPR (clustered regularly interspaced short palindromic repeats) screening is commonly used to identify potential targets in cellular or animal disease models.

Illustrated example: applications in lung cancer.



Shen, Wanying, Fangli Hu, Pan Lei, and Yijun Tang. "Applications of CRISPR Screening to Lung Cancer Treatment." Frontiers in Cell and Developmental Biology 11 (December 2023). https://doi.org/10.3389/fcell.2023.1295555.

Alternative: chemical library screening



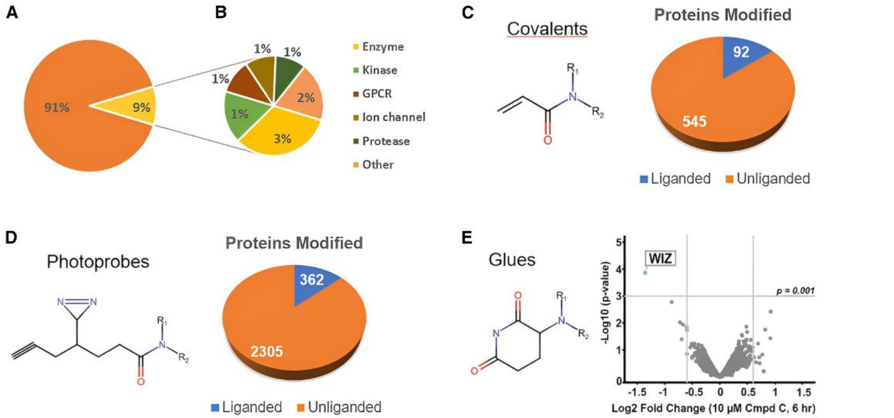
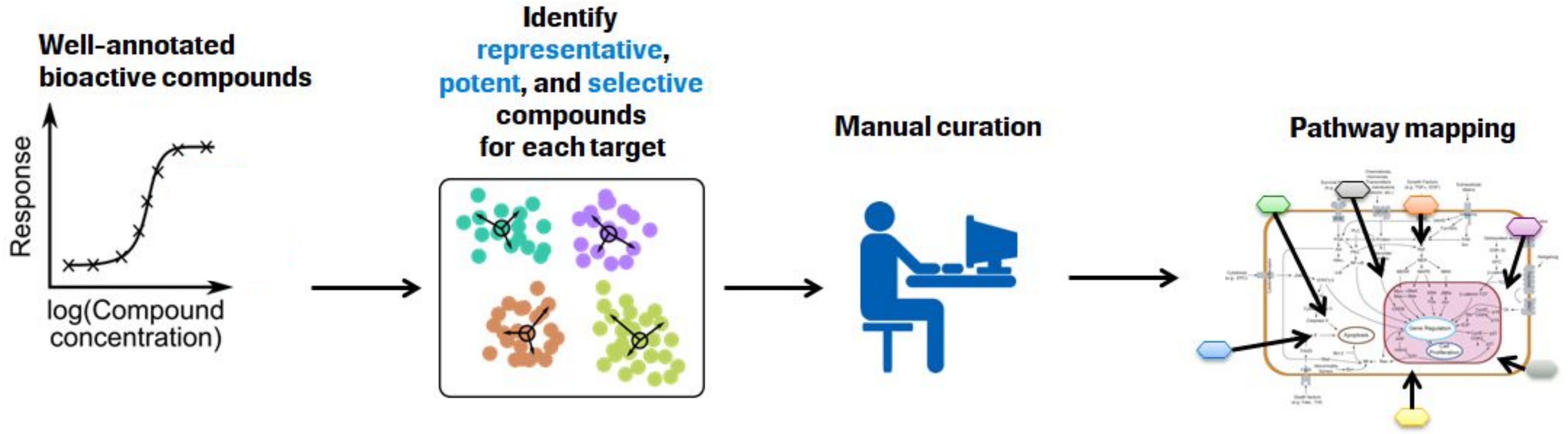


Figure 1. Genome coverage of compound libraries
(A) Coverage of the most extensive chemogenomics library published to date as a fraction of the human genome and (B) target classes covered in the library; adapted from Canham et al.[8] Expanding biological space through new types of compounds including (C) covalent (liganded status based on DrugBank), (D) photoreactive (liganded status based on T clin + T chem vs. T bio + T dark),[9] and (E) molecular glues; adapted and reproduced with permission from Backus et al.,[10] Offensperger et al.,[11] and Ting.[12]

Left: Oprea, Tudor I., Cristian G. Bologa, Søren Brunak, Allen Campbell, Gregory N. Gan, Anna Gaulton, Shawn M. Gomez, et al. 2018. "Unexplored Therapeutic Opportunities in the Human Genome." Nature Reviews Drug Discovery 17 (February): 317–32. https://doi.org/10.1038/nrd.2018.14; Right: Vincent, Fabien, and Davide Gianni. "The Limitations of Small Molecule and Genetic Screening in Phenotypic Drug Discovery." Cell Chemical Biology (2025). https://doi.org/10.1016/j.chembiol.2025.10.008.

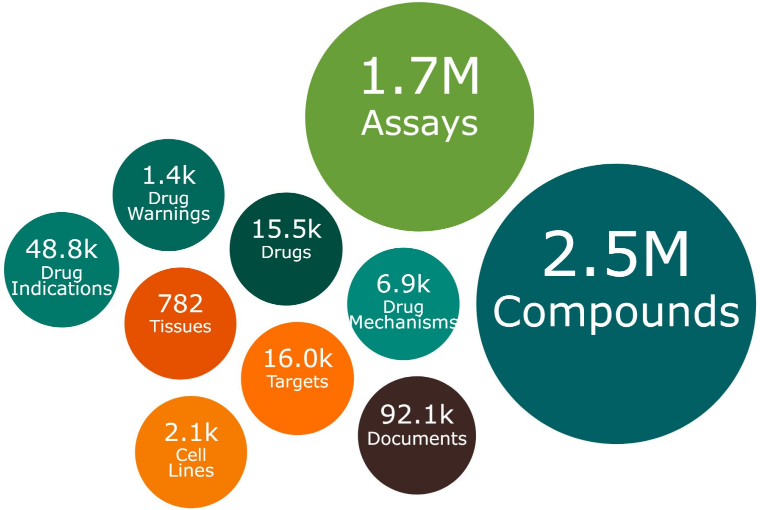
The Small-molecule PAthway Research Kit (SPARK)

Now known as the Pathway Annotated Chemical Ensemble (PACE) *library*



The ChEMBL database

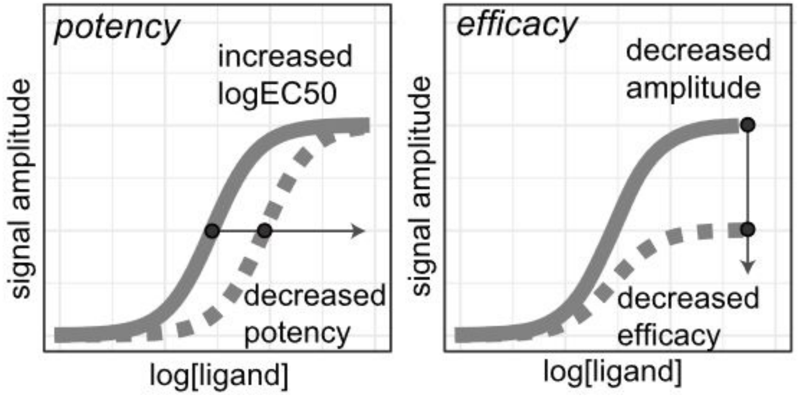
- An example of query: aspirin.
- Systematic and programmatic accession via ChEMBL API (source code).
- We can use **dose-response data** to annotate the *triplets* of compound, assay activity, and targets.



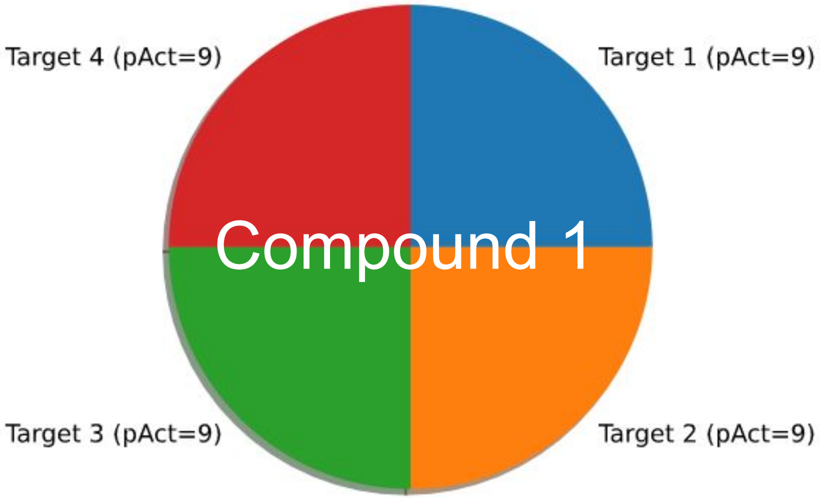
Visualization of ChEMBL
(version 35; Dec 2024)

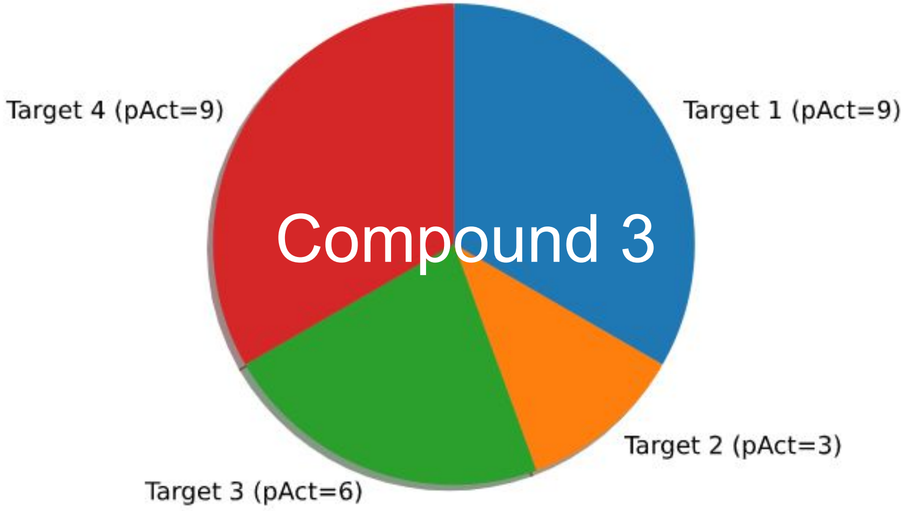
Discussion

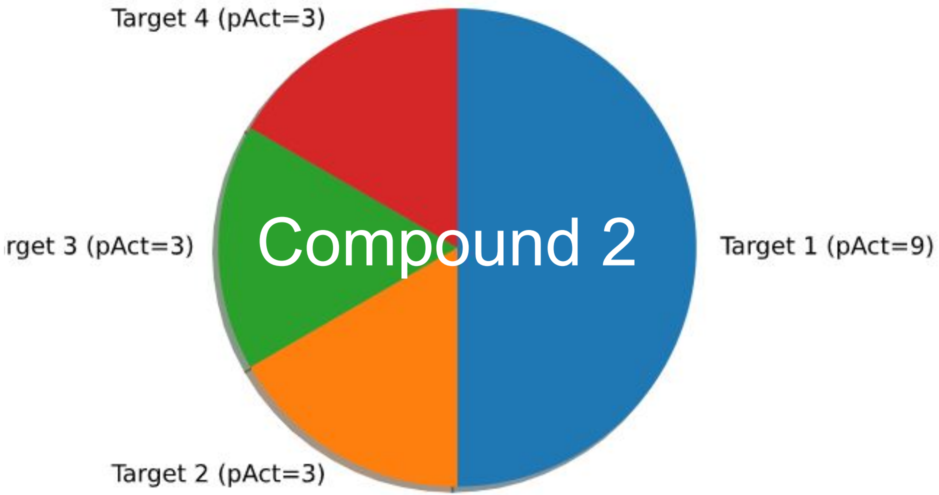
1. Why do we care selecting *representative*, *potent*, and *selective* compounds?
2. How to define following terms mathematically ...
   a. Representativity?
   b. Potency?
   c. Selectivity?



A toy example about how to quantify a compound’s potency and selectivity

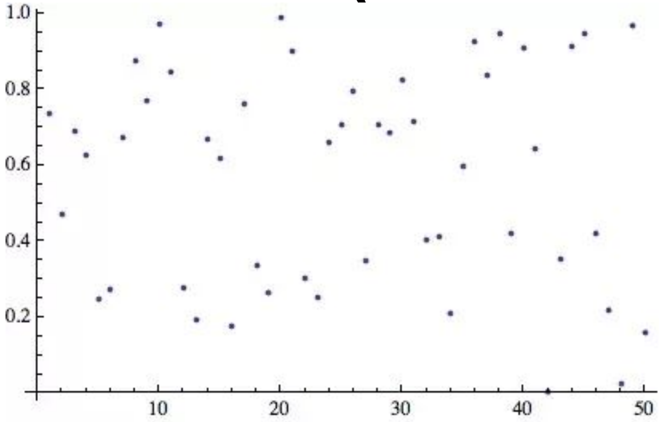




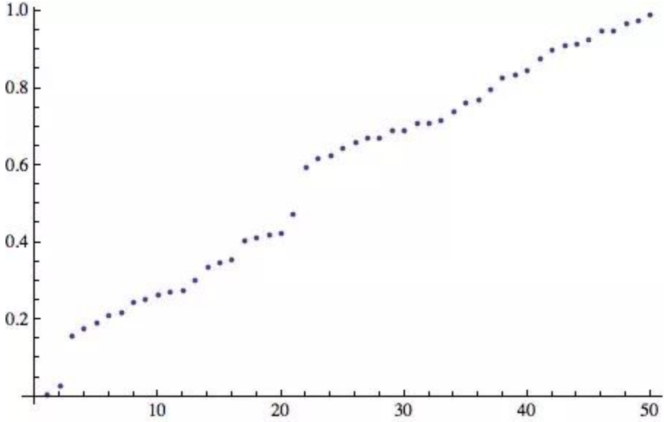


The Gini Index (a.k.a. Gini Coefficient)

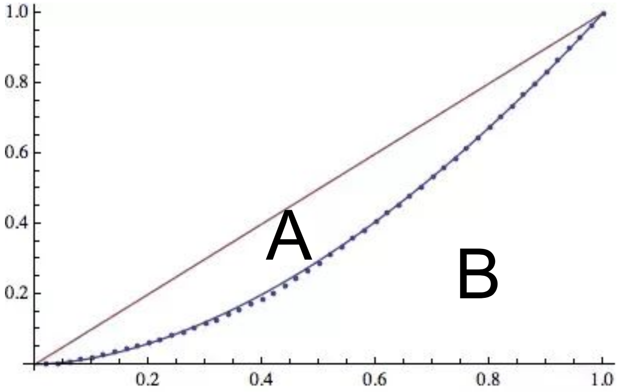
A random vector of 50 values



Sorted from low to high



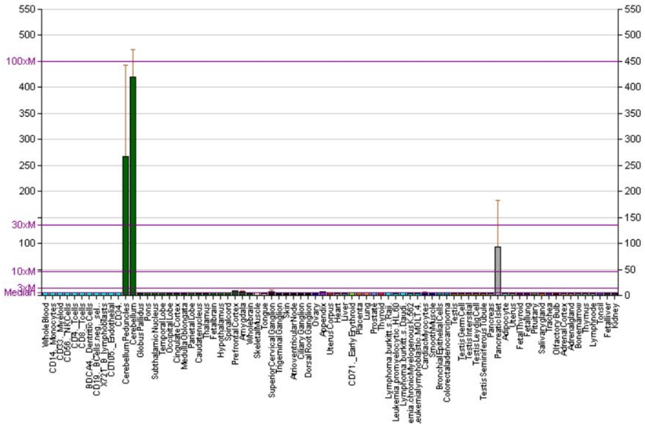
The Gini Index is calculated based on the cumulative distribution: larger value→large inequality



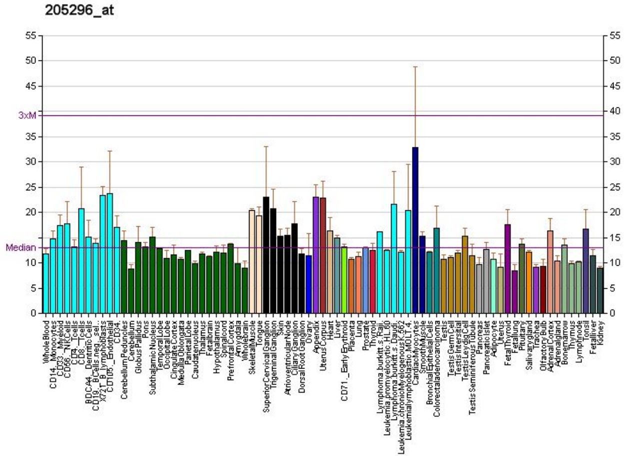
$G = A/(A+B)$

The Gini Index quantifies inequality/ selectivity

NEUROD1

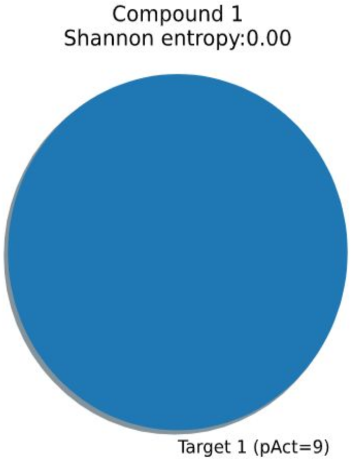


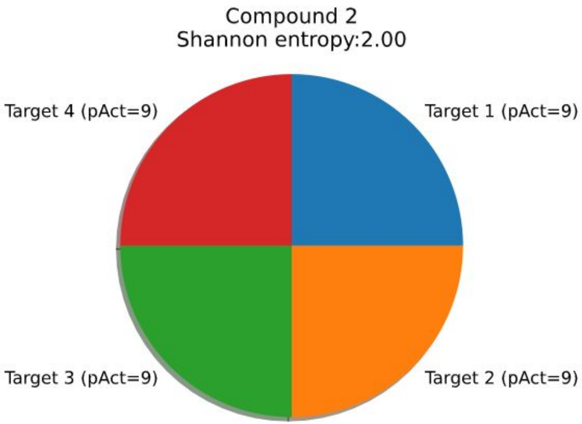
RBL1

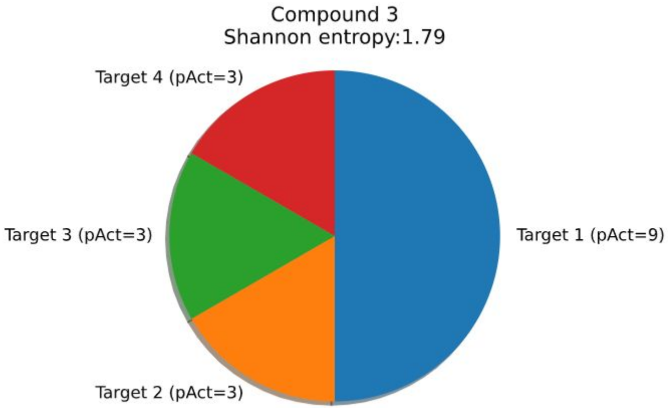


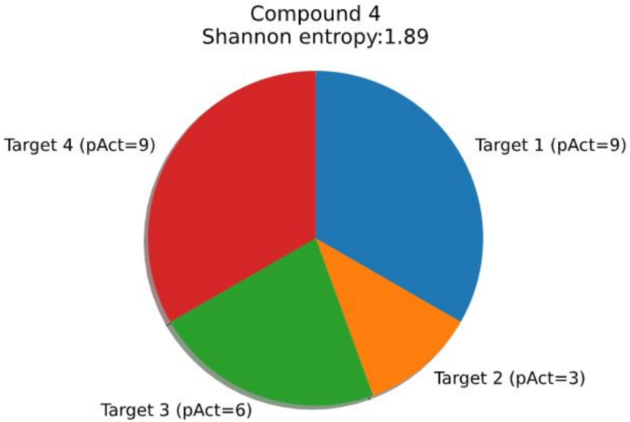
The Gini Index of expression of *NEUROD1* across tissues is near 1, whereas that of *RBL1* is near 0.

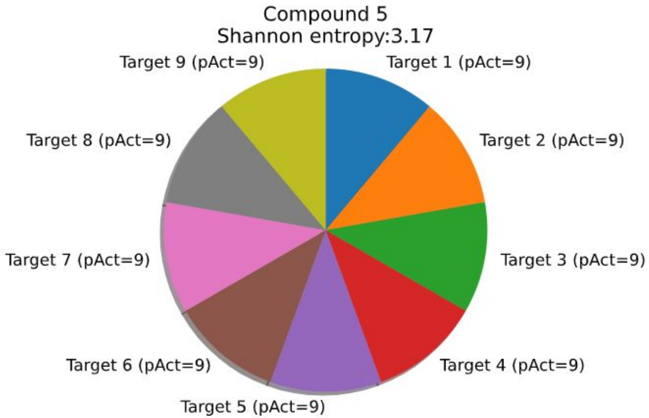
An alternative metric: Shannon's Entropy





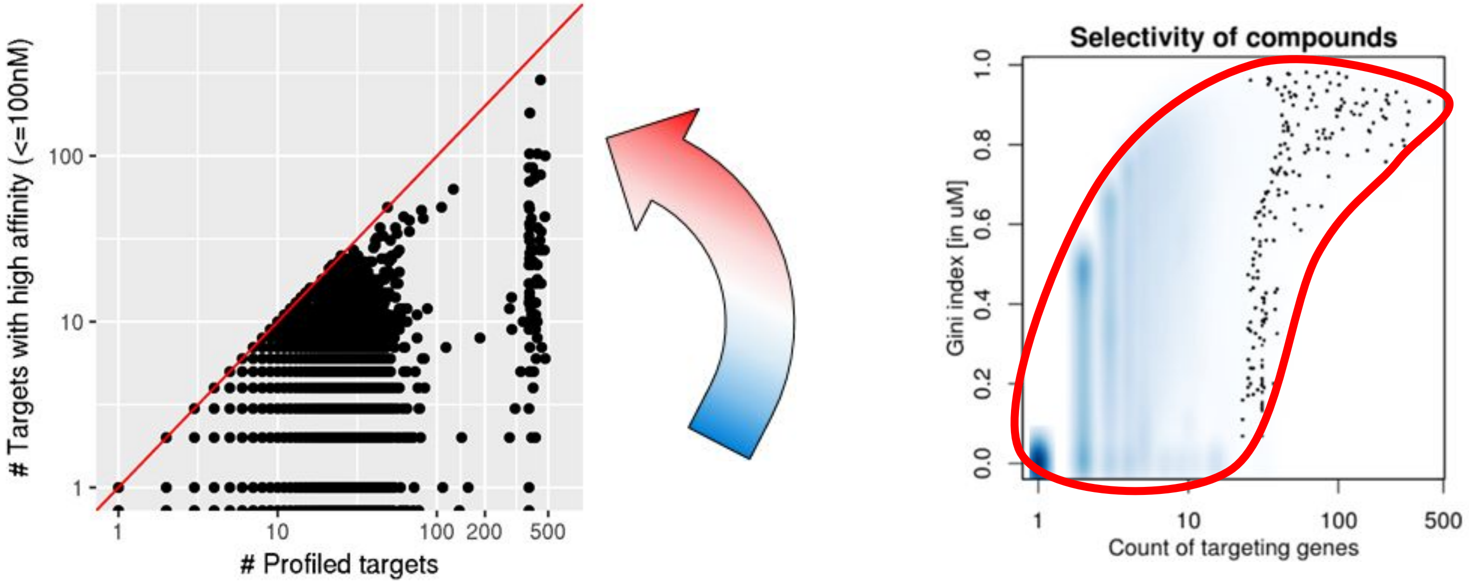






$$\mathrm{H}(X) := -\sum_{x \in \mathcal{X}} p(x) \log p(x)$$

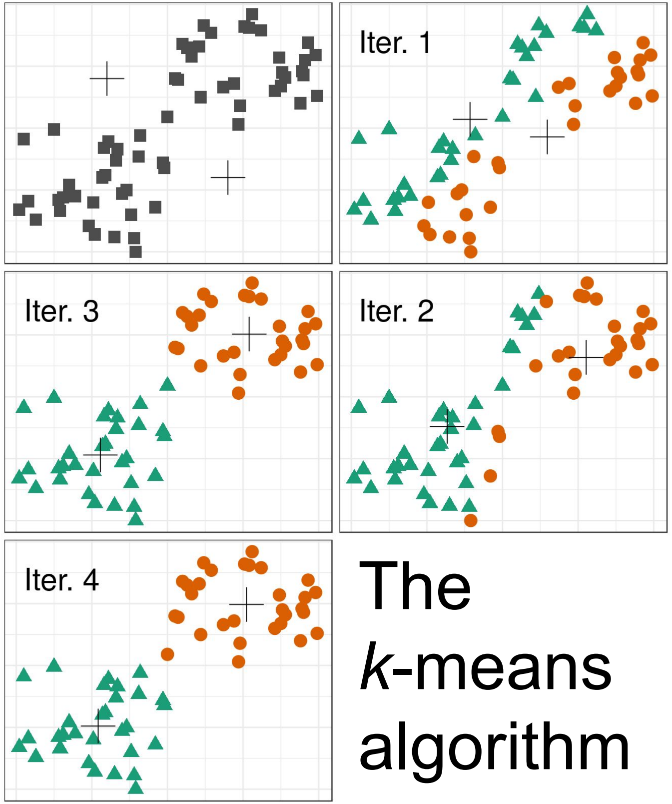
Count of targets and selectivity of ChEMBL molecules



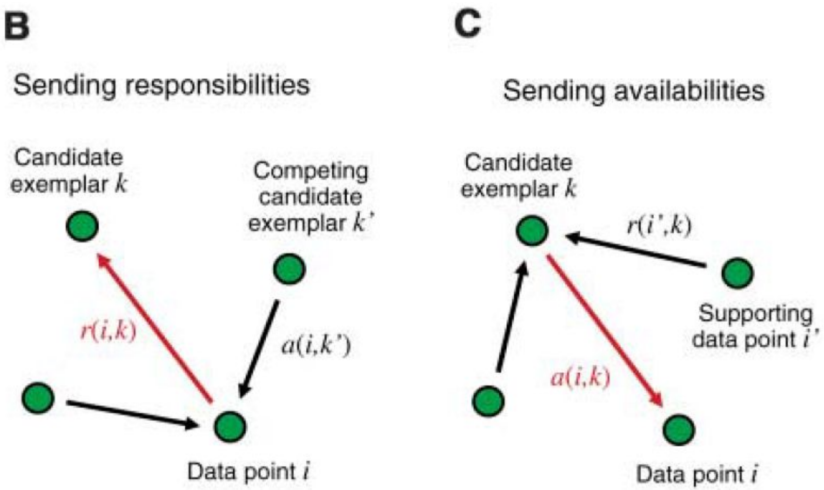
With some exceptions, most compounds are profiled against <100 targets. We distinguish between specific and pleiotropic compounds.

The **shark-fin shape** curve suggests that frequently profiled compounds tend to be more selective (and *vice versa*).

Unsupervised clustering

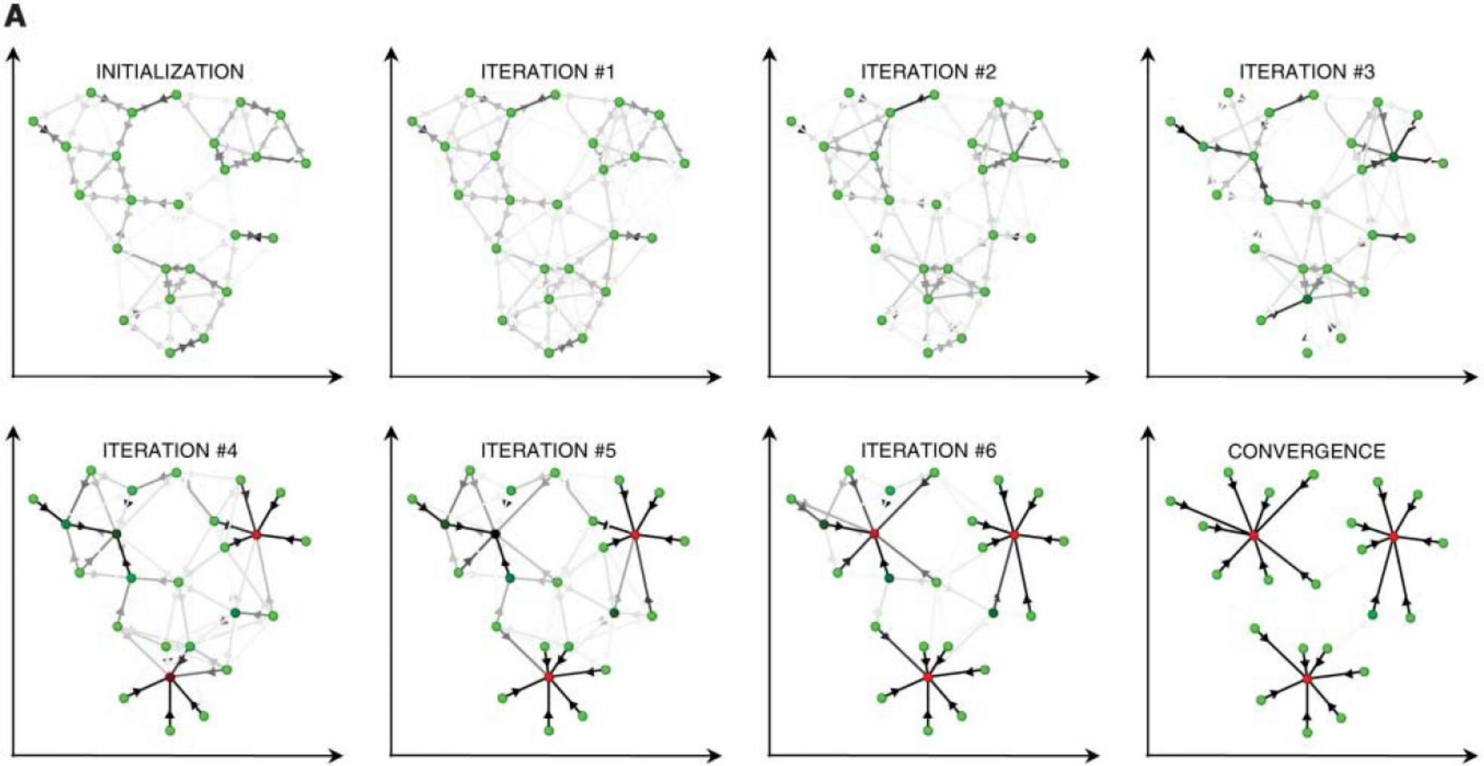


The
k-means
algorithm



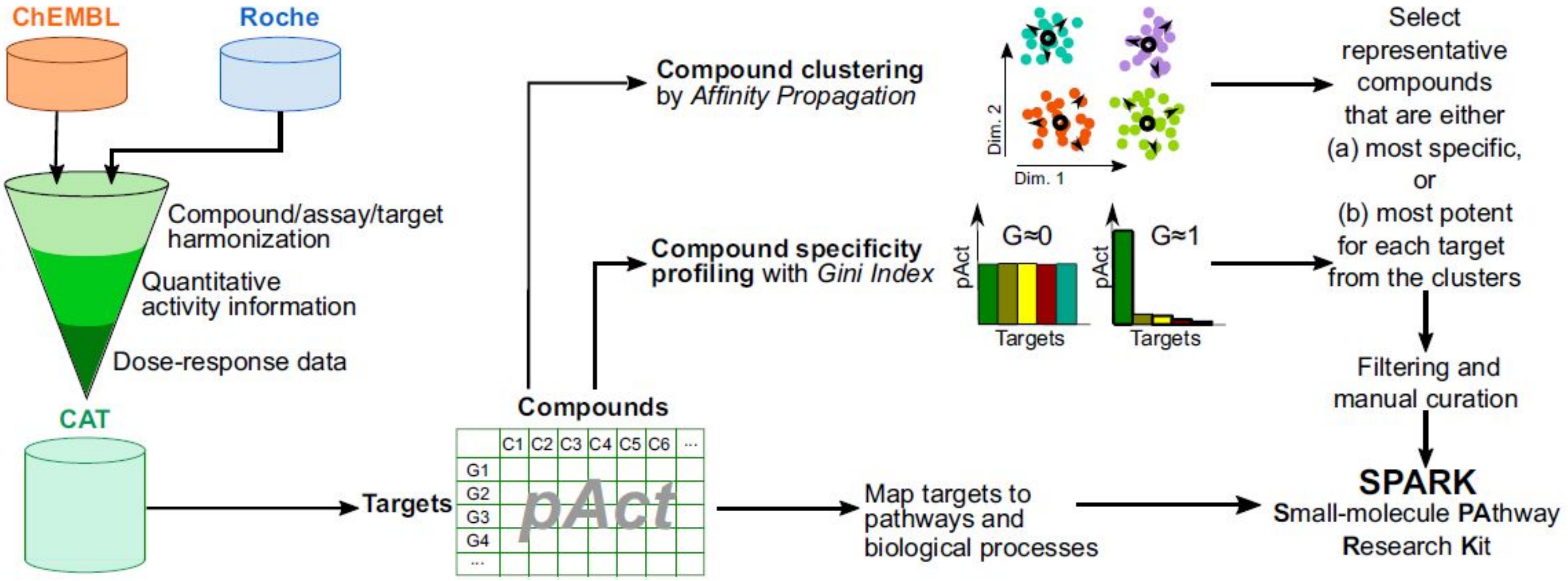
Affinity Propagation updates **responsibilities** and **availabilities** iteratively

Affinity Propagation in action



[A movie of iterations]

Construction of SPARK in detail



Harmonization

… of public and Roche internal data

Machine learning

… to select compounds

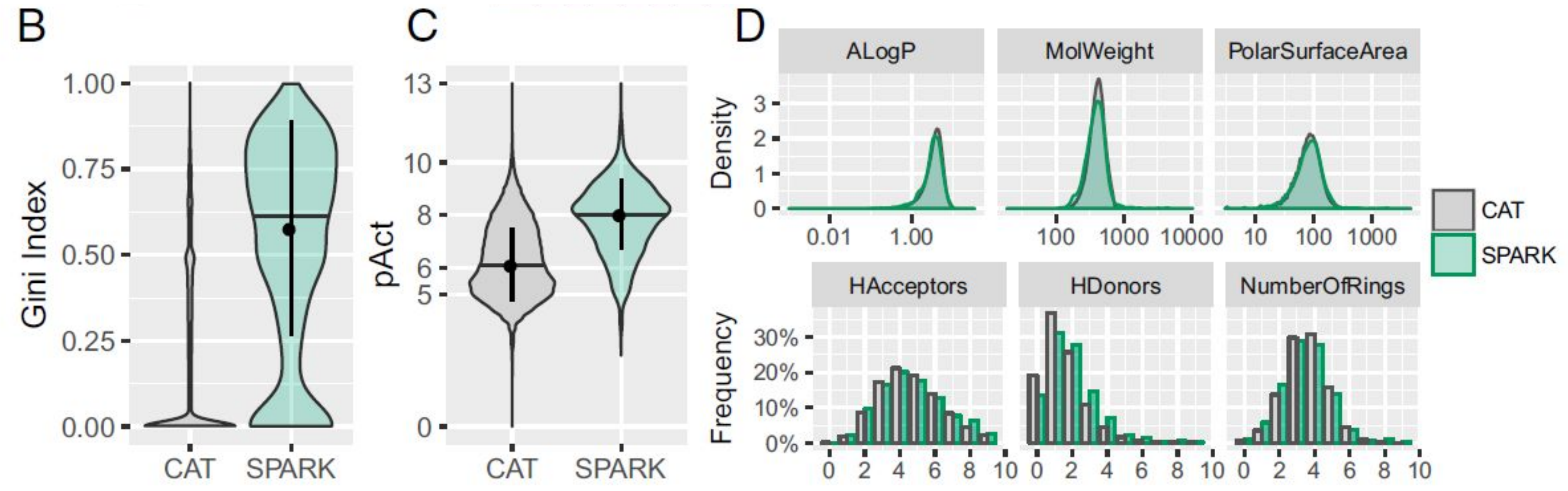
Pathways

… mapped to compounds

Curation

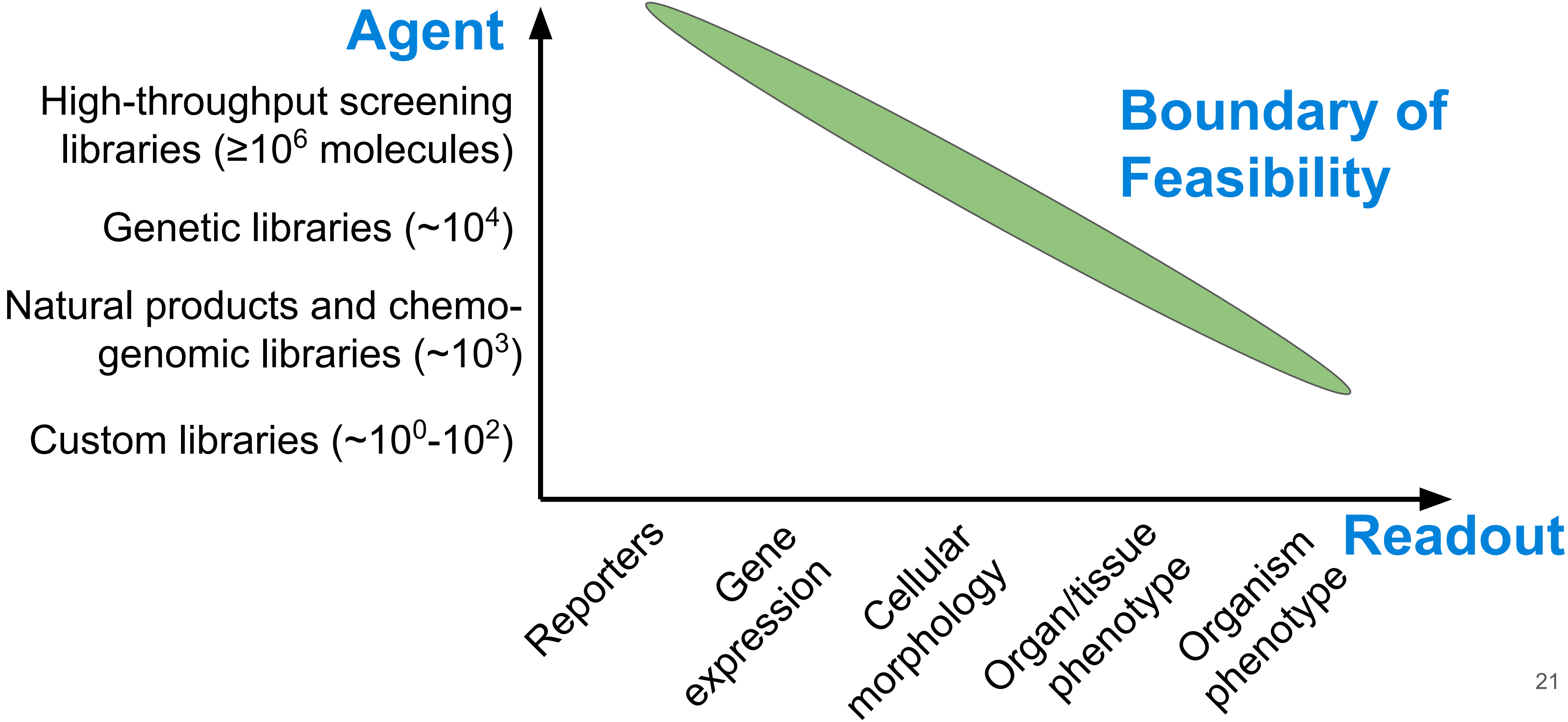
… to enrich quality compounds

SPARK covers the chemical space evenly with representative, potent, and specific compounds



Roudnicky *et al.*, PNAS, 2020, https://www.pnas.org/content/early/2020/08/04/1911532117

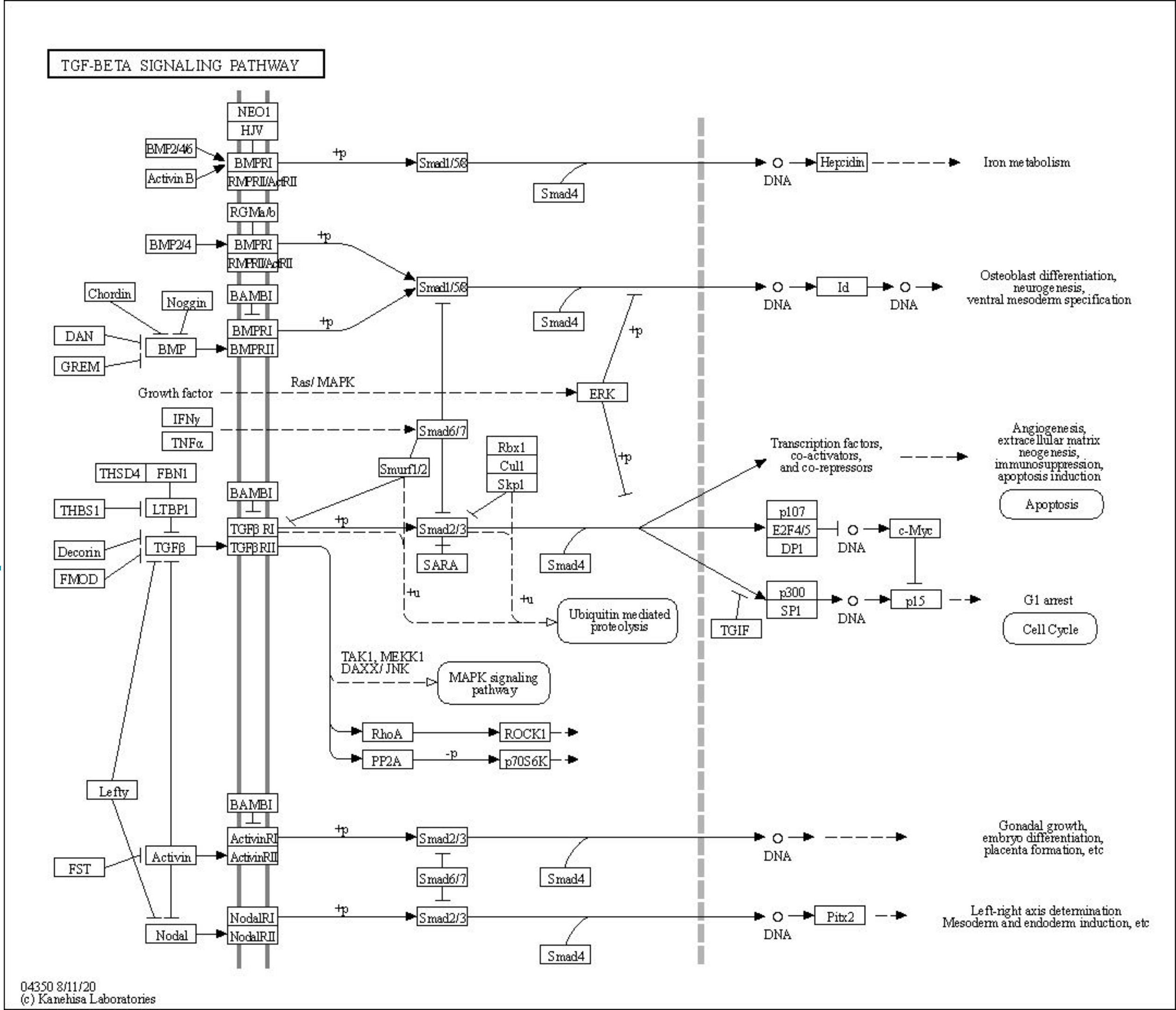
Phenotypic screenings by agent and readout



Mapping genes to biological pathways

Option 1: [KEGG pathways](), with the example of [TGF-β signaling pathway]().

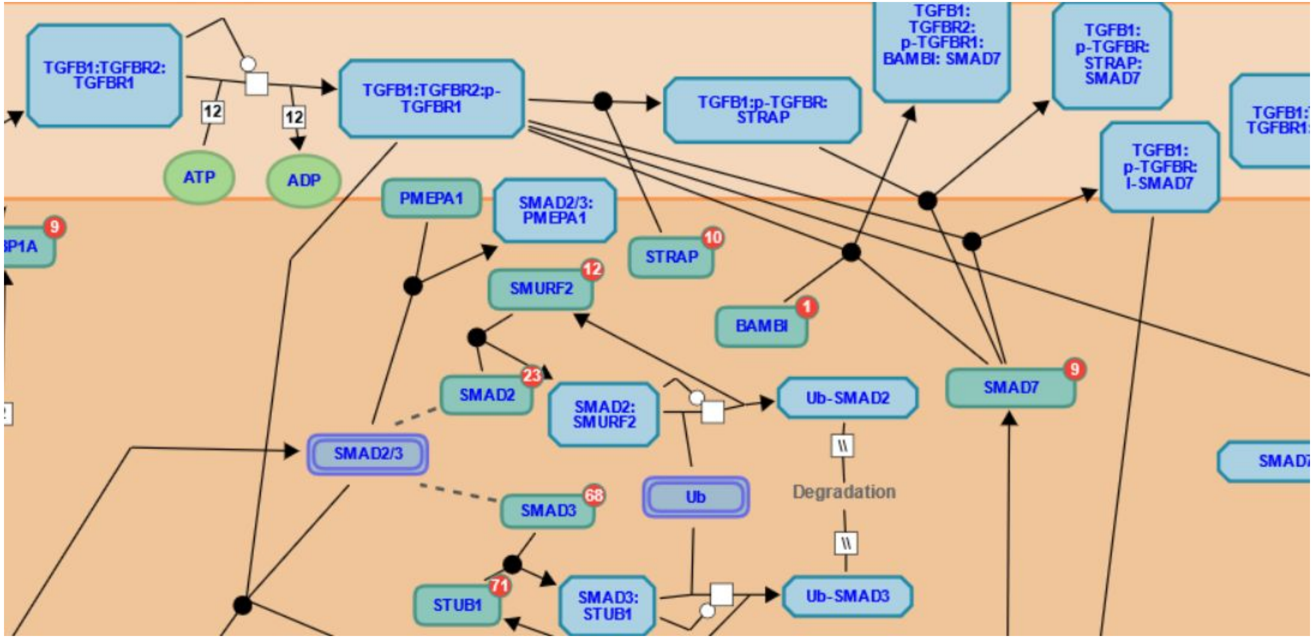
[A RESTful API]() is available for academic use, with clients in Python and R.



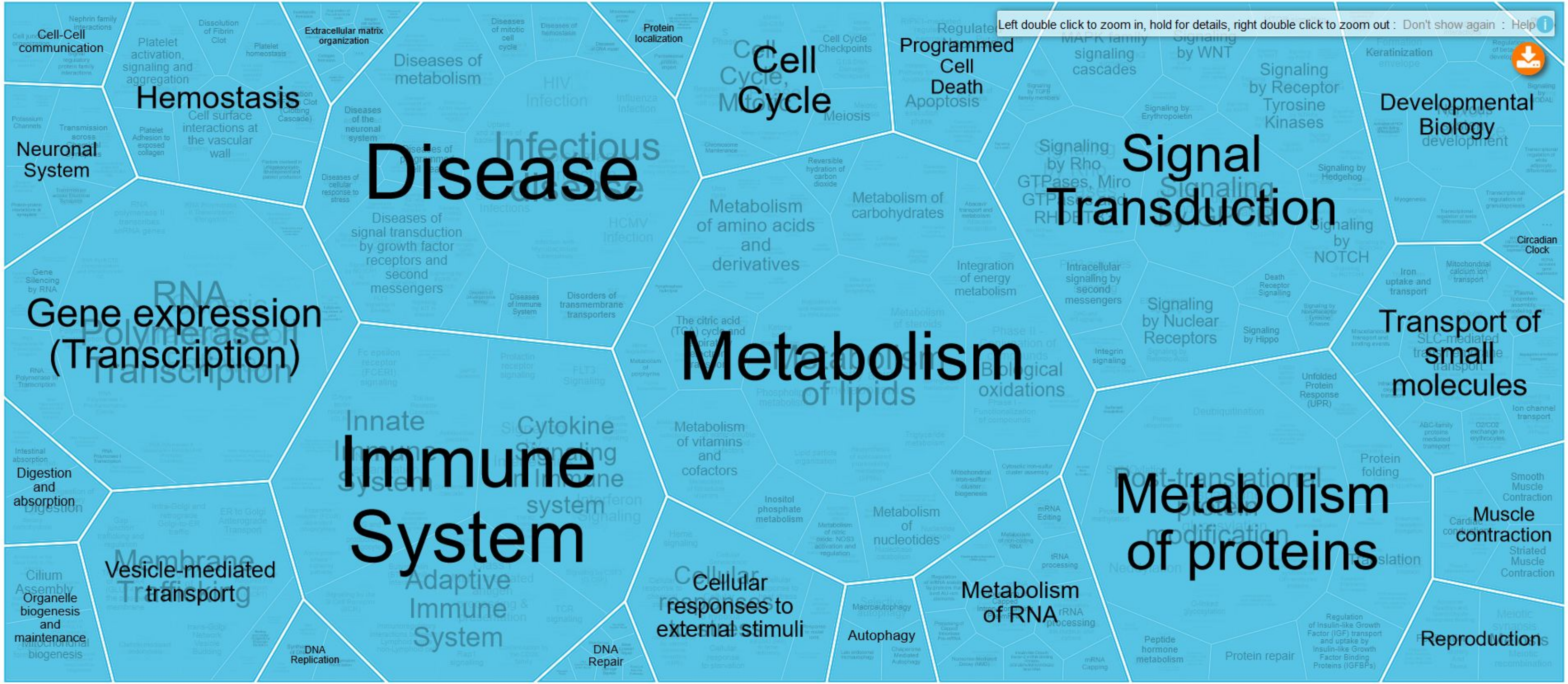
Mapping genes to biological pathways

Option 2: Reactome pathways, with the example of the TGF-β signaling pathway.

Developer's Zone provides API and graph database interfaces.



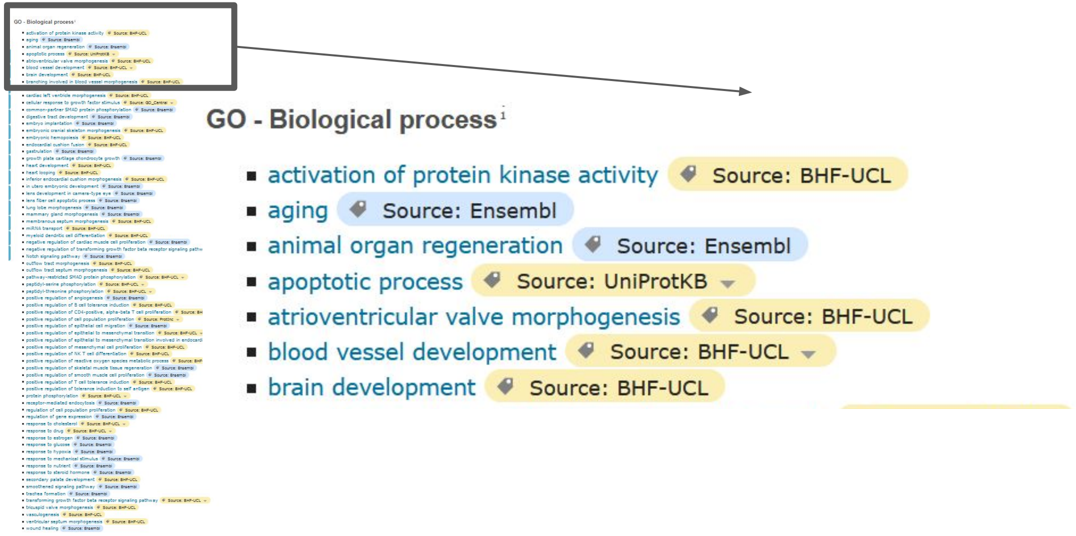
Overview of pathways captured by Reactome



The Voronoi (Reacfoam) view of all pathways in Reactome

Mapping genes to biological processes

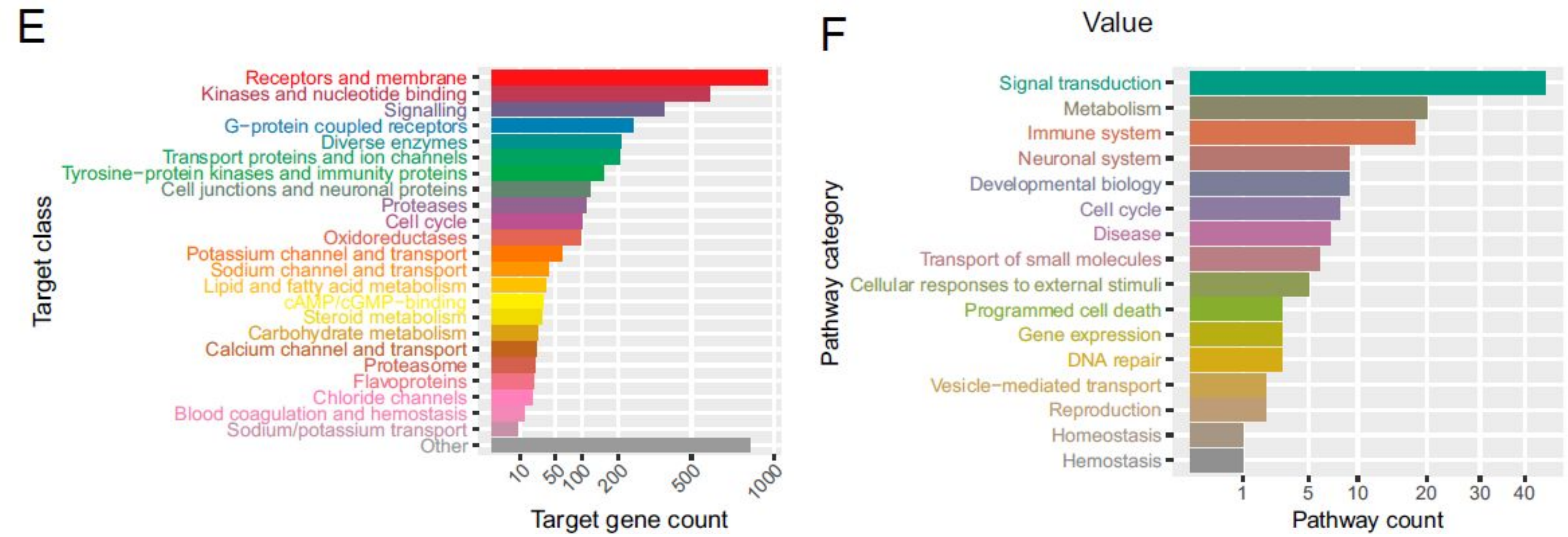
- [Gene Ontology]
- [UniProtKB keywords]
- Example: TGFBR2_HUMAN (TGF-beta receptor type -2, P37173)



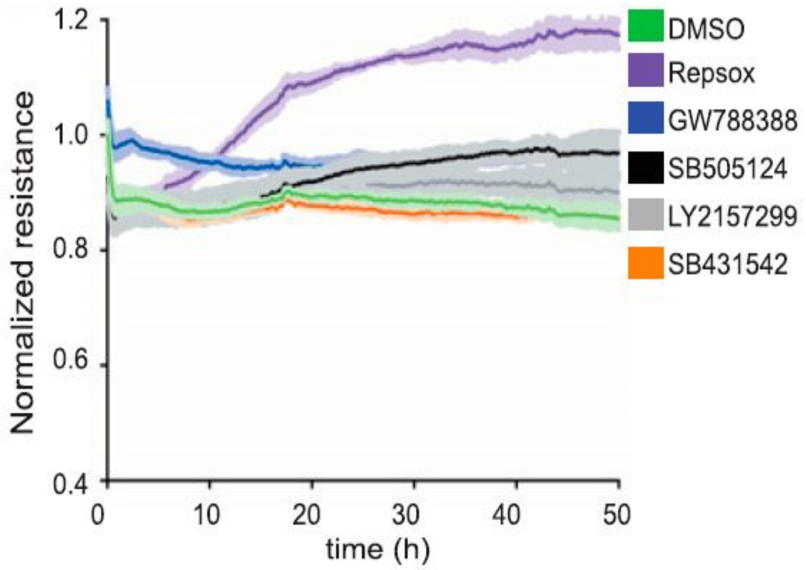
Keywords

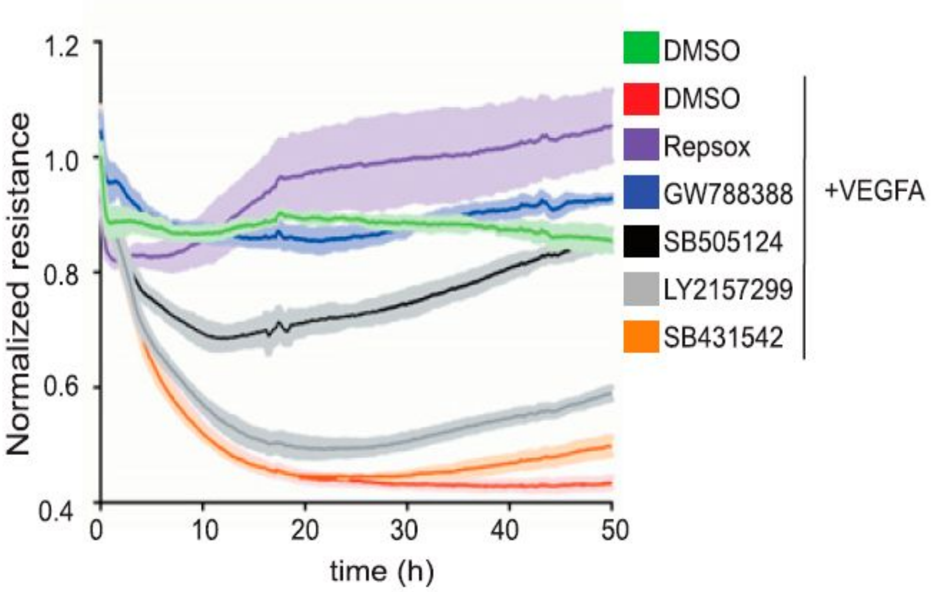
Molecular function	Kinase, Receptor, Serine/threonine-protein kinase, Transferase
Biological process	Apoptosis, Differentiation, Growth regulation
Ligand	ATP-binding, Magnesium, Manganese, Metal-binding, Nucleotide-binding

SPARK covers the target space evenly with representative, potent, and specific compounds



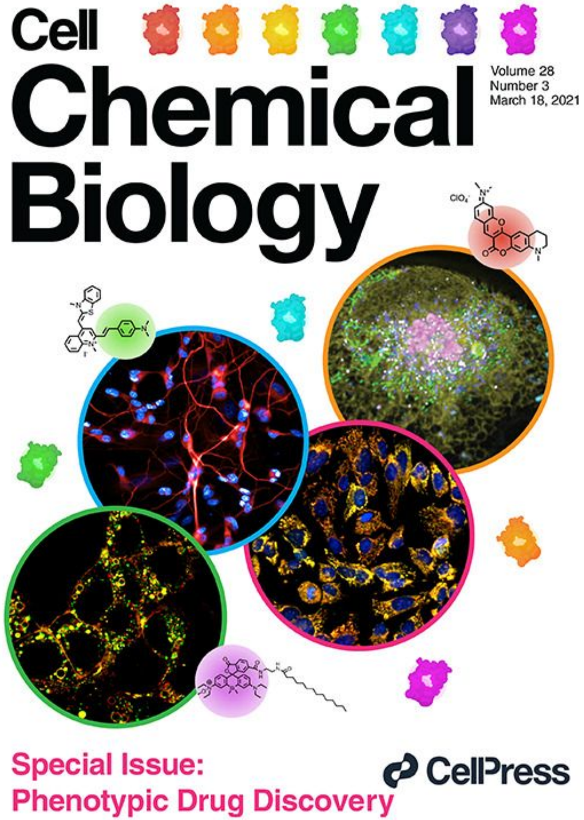
Screening with SPARK in endothelial cells identified TGF-β pathway genes as potential targets for diabetic retinopathy





Conclusions about chemogenomic library

- Phenotypic drug discovery can lead to first-in-class drugs with novel mechanisms;
- Unsupervised machine learning and data modelling contribute to build chemogenomic libraries;
- We can link drug candidates via targets to biological pathways and processes.



The limitations of genetic screening in phenotypic drug discovery

Table 3. Genetic screening limitations and mitigation strategies

Genetic screening limitations	Mitigation strategy
Effects have a delayed onset	Extend assay duration
Effects may not capture allosteric or scaffolding function of a protein	Use genetic libraries that have well-annotated small molecule tool compounds or antibody/protein tools, development of tunable KD protocols,
Genetic redundancy	Develop combinatorial gene editing protocols to examine contribution of multiple genes to a phenotype
Workflows for genome-wide screens are not compatible with use of translatable cellular models of disease	Use shorter and focused gene lists leveraging human genetics, omics, literature, and AI predictions
Resource and time intensive	Use of "generic" assay readouts (e.g., cell painting, transcriptomics)
Focus on gene KD/KO may miss activation mechanisms	Additional CRISPRa or cDNA screen
May miss selective modulation via macromolecular complexes	N/A

KD, knockdown; KO, knockout; N/A, not applicable.

Vincent, Fabien, and Davide Gianni. "The Limitations of Small Molecule and Genetic Screening in Phenotypic Drug Discovery." Cell Chemical Biology (2025). https://doi.org/10.1016/j.chembiol.2025.10.008.

The limitations of small molecule screening in phenotypic drug discovery

Table 2. Small molecule screening limitations and mitigation strategies

Small molecule screening limitations	Mitigation strategies
Chemogenomics libraries only cover ~5%–10% of the genome	Additional genome coverage via secreted proteins or receptor antibodies
Large legacy compound libraries only cover a fraction of the genome	New compound classes address a larger fraction of the genome (covalent, photoactivatable, and molecular glue compounds)
Limited throughput for complex assays, especially *in vivo* assays	Screening library compression, smaller compounds (e.g., fragments), biological diversity-driven libraries
Frequent lack of selectivity for related protein homologs	New compound classes may offer greater selectivity (covalent and molecular glue compounds)
Potential species translation hurdle	Additional species phenotypic assays, 10× *in vivo* dosing strategy, complex human models
Target and MoA deconvolution post-screen often challenging and may be required for successful outcome	New compound classes (covalent, photoactivatable, and glue) may facilitate target and MoA deconvolution through proteomics

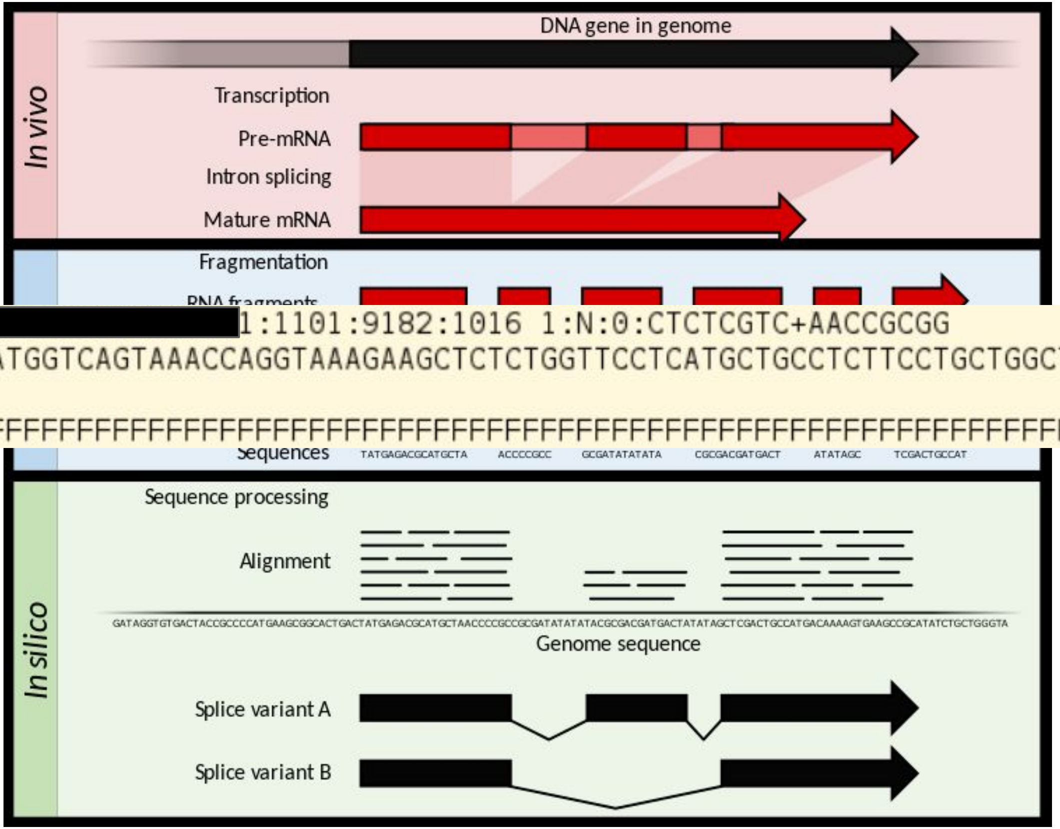
Vincent, Fabien, and Davide Gianni. “The Limitations of Small Molecule and Genetic Screening in Phenotypic Drug Discovery.” Cell Chemical Biology (2025). https://doi.org/10.1016/j.chembiol.2025.10.008.

Offline activities of Module II

Please use your favourite programming language (shell scripts, python, R, for instance) and APIs (Application Programming Interfaces) of databases to perform following operations. Submit your code.

1. Retrieve all approved drugs from the ChEMBL database, sort them by approval year and name (a Python example is here; documentations of the ChEMBL API can be found here);
2. For each approved drug **since 2019** that you identified in step (1), retrieve a list of UniProt accession numbers, namely protein targets associated with the drug;
3. For each protein with a UniProt accession number that you identified in step (2), retrieve UniProt keywords associated with it. You can use the UniProt API, documented here. Python and R clients are also available.

Transcriptome profiling by RNA sequencing

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@[redacted]1:1101:9182:1016 1:N:0:CTCTCGTC+AACCGCGG
GNATCAGAATGGTCAGTAAACCAGGTAAAGAAGCTCTCTGGTTCCTCATGCTGCCTCTTCCTGCTGGCTTTATT
+
F#FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
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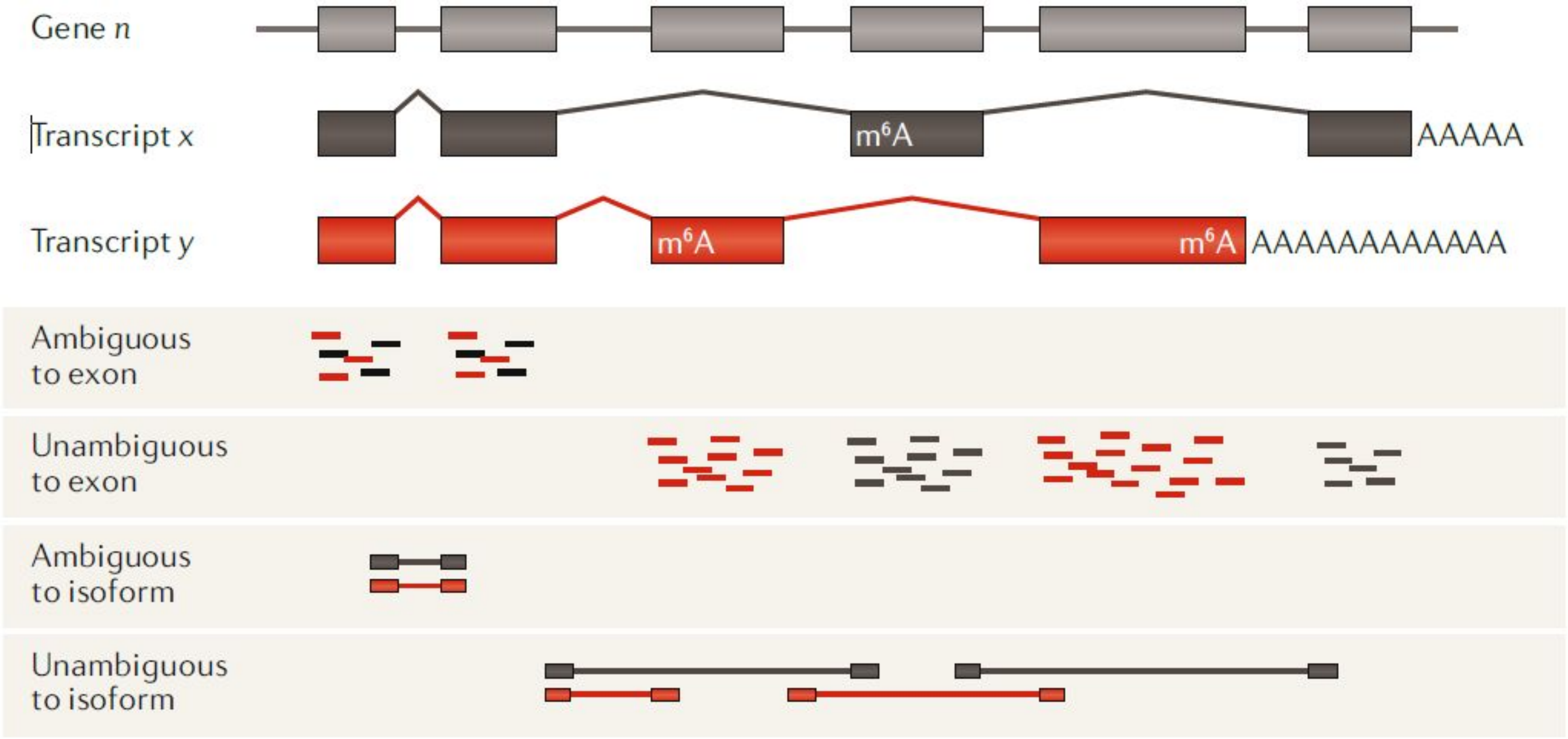
Transcriptome profiling by RNA sequencing



Differential gene expression

Read Mapping

Count collection

	sample A1	sample A2	sample B1	sample B2
gene 1	8	10	100	200
gene 2	14	15	15	40
gene 3	33	40	35	70
...	...	...	...	...
gene N	100	120	105	220

Normalization by library size

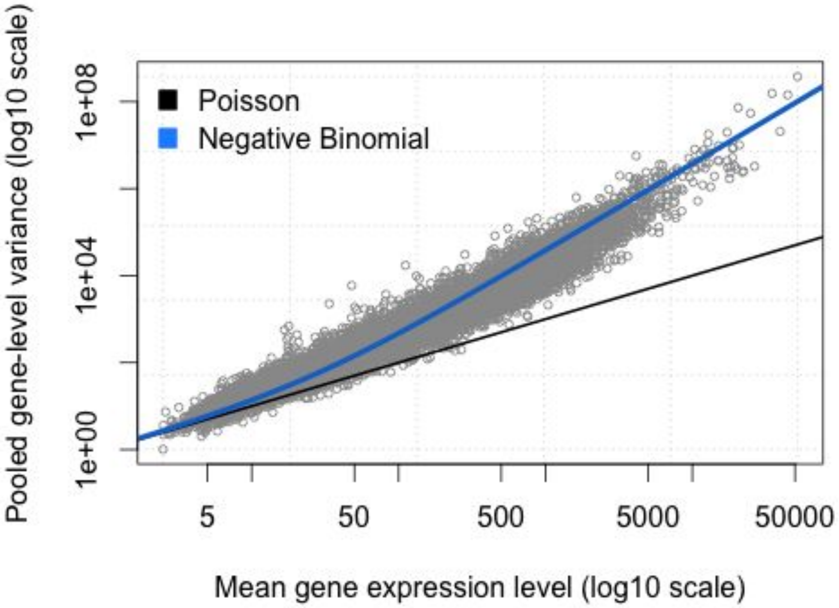
	sample A1	sample A2	sample B1	sample B2
gene 1	8	10	100	200
gene 2	14	15	115	40
gene 3	33	40	35	70
...	...	...	...	...
gene N	100	120	105	220

Tot. reads: 5 millions

Tot. reads: 10 millions

Differential Gene Expression Analysis

	sample A1	sample A2	sample B1	sample B2
gene 1	0.16	0.20	2.00	2.00
gene 2	0.28	0.30	0.30	0.40
gene 3	0.66	0.80	0.70	0.70
...	...	...	...	...
gene N	2.00	2.40	2.10	2.20



Tools: *edgeR* and *DESeq2*

Probability theory and statistical tools discussed

- Distributions
  - Gaussian distribution (used in linear model)
  - Bernoulli distribution → Binomial distribution → Negative binomial distribution
  - Poisson distribution → Negative binomial distribution
  - Poisson distribution ←→ Exponential distribution
- Statistical methods
  - Bootstrapping method
  - Student’s t-test
  - Wilcoxon-Mann-Whitney test
  - Kolmogorov-Smirnov test

Interpret differential gene expression data with gene-set enrichment analysis

Gene ($N \sim 10^4$)	G_1	G_2	G_3	G_4	G_5	...	G_{N-3}	G_{N-2}	G_{N-1}	G_N
Change (log2FC)	3.0	2.8	2.5	1.5	1.2	...	-0.8	-1.2	-1.5	-2.2

Differential gene expression results

Reactome pathways

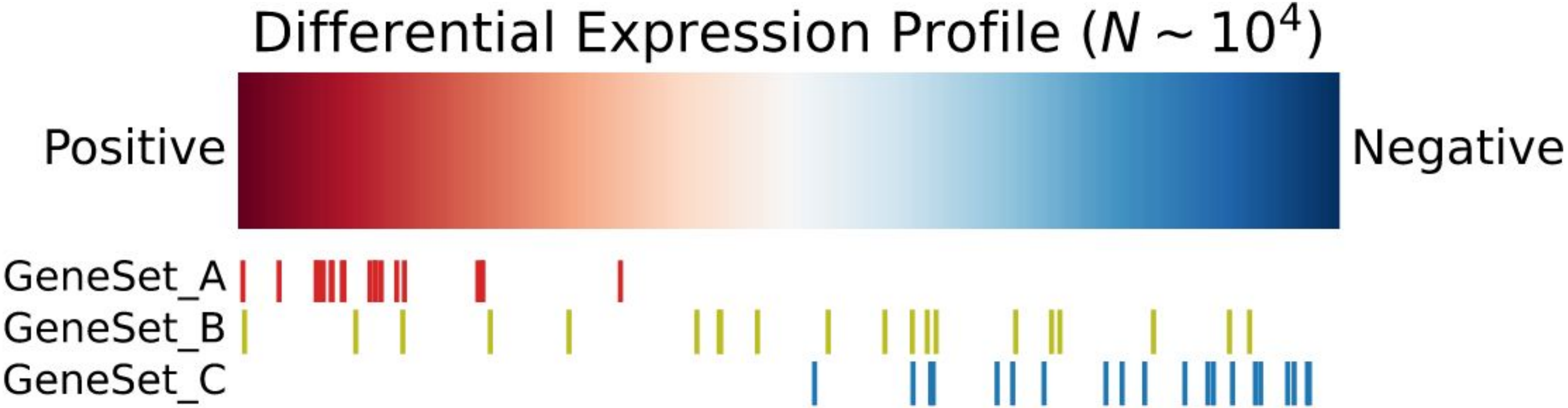
Gene Ontology

UniProt Keywords

Literature

Gene-set Enrichment Analysis Methods

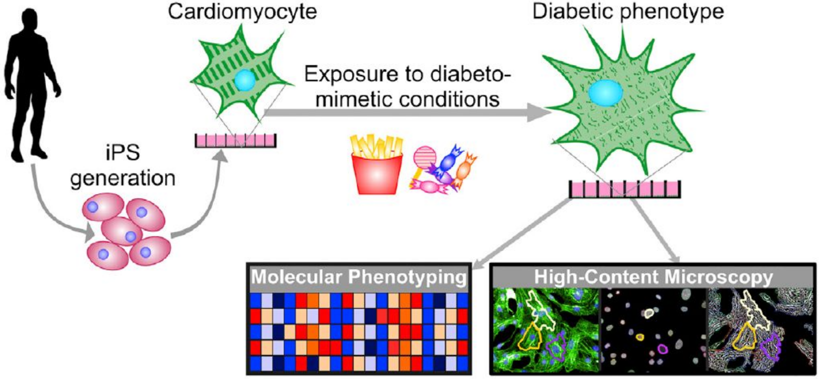
Gene-set enrichment analysis

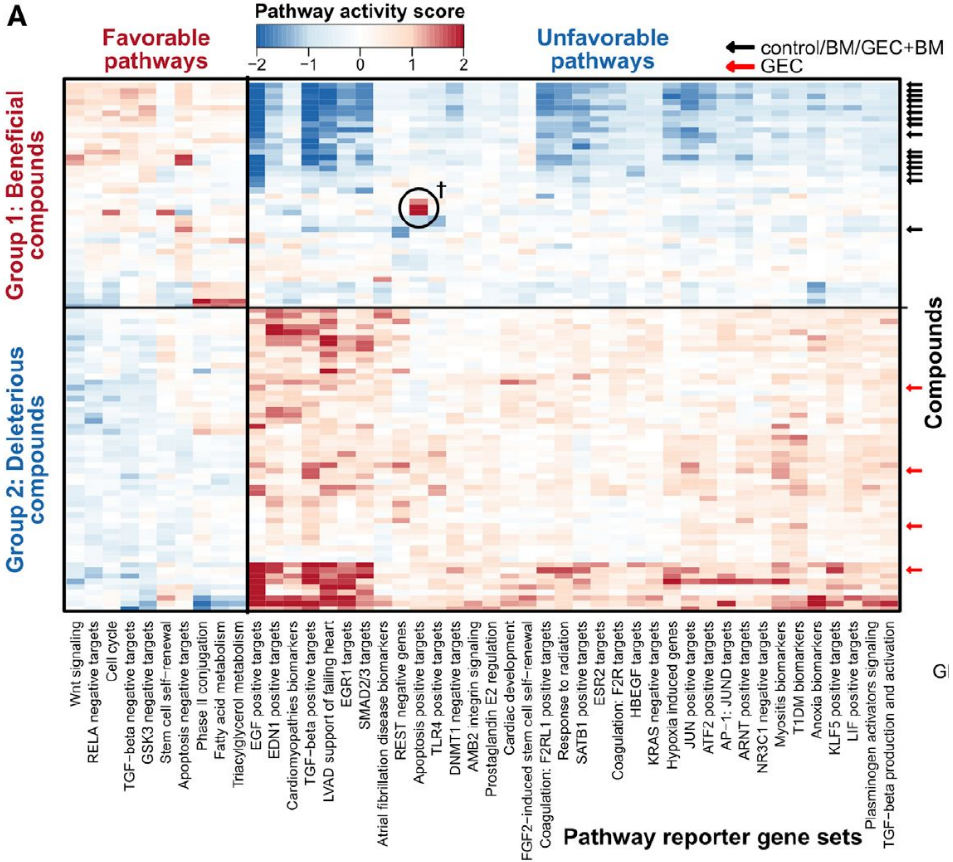


Input: (1) a differential gene expression profile; (2) a set of gene-sets {*G*}, each a set of genes.

Output: a ranked list of the input gene-sets by *enrichment*.

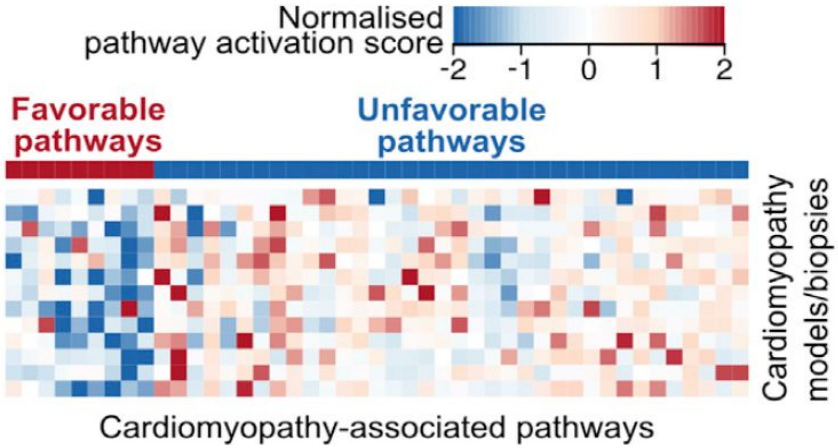
Gene expression as screening readout

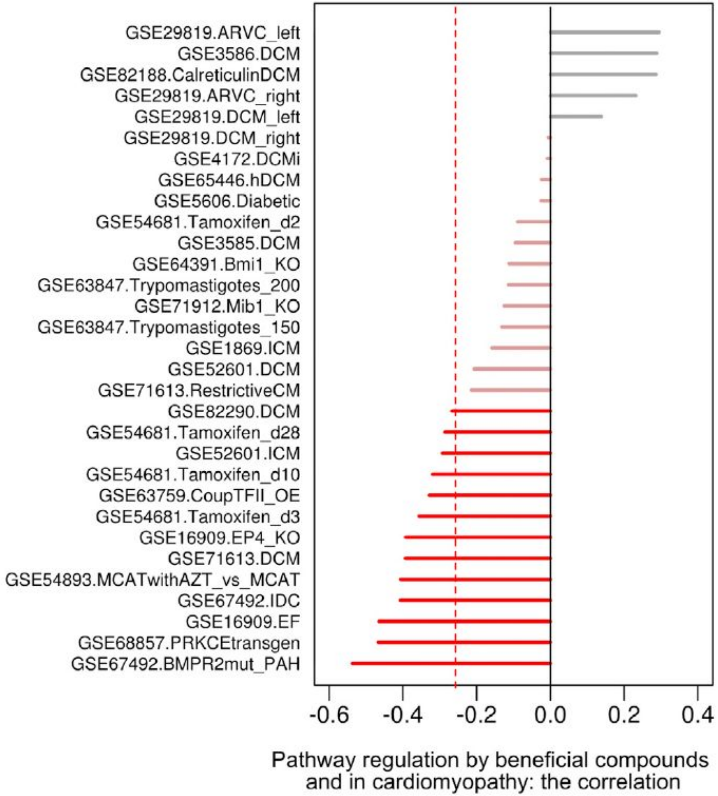




Differential gene expression profiles are molecular snapshots of drugs' action in the cell.

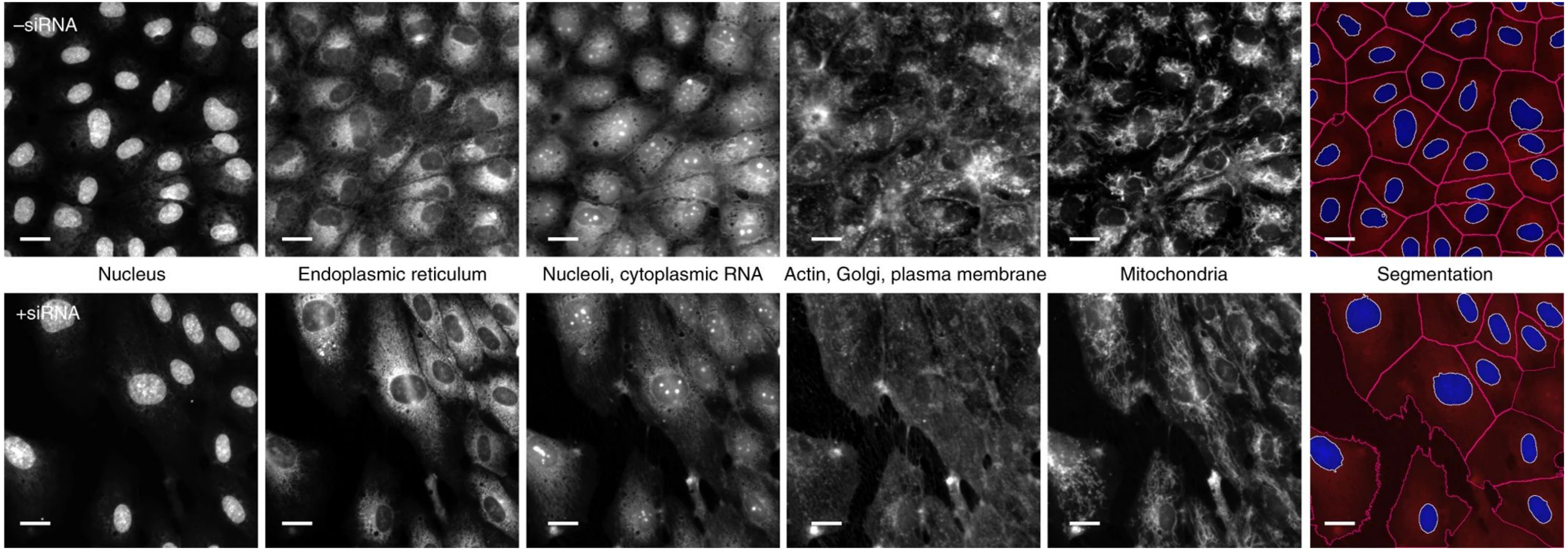
Gene expression from patient and animal models help compound selection

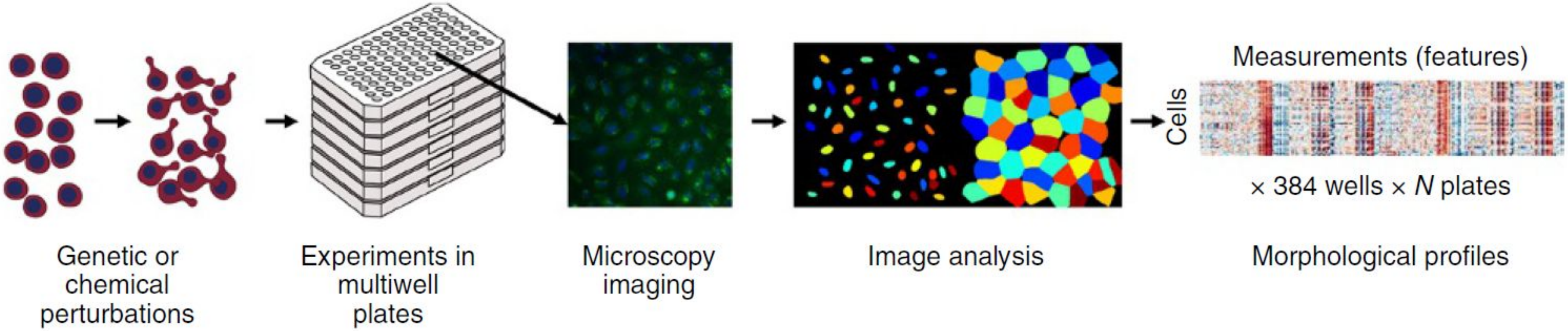




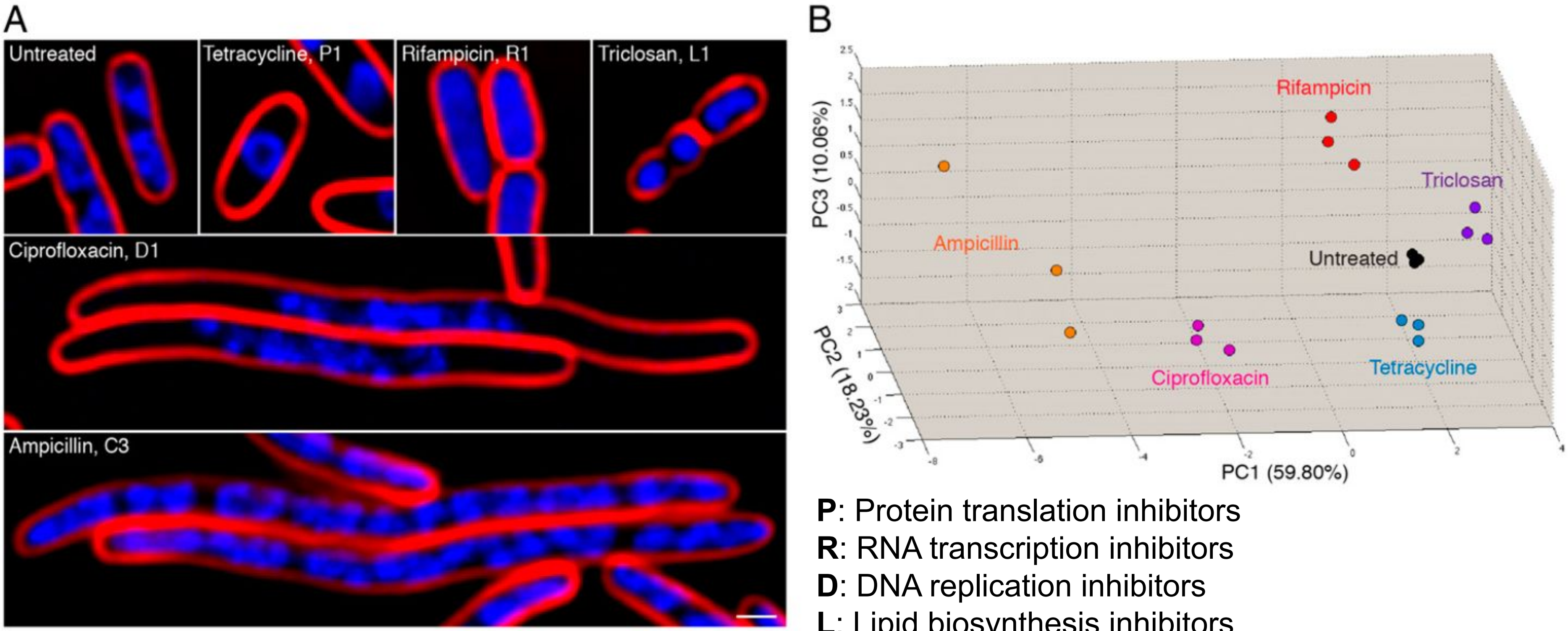
We can prioritise molecules that reverse disease-induced changes.

Morphology as screening readout



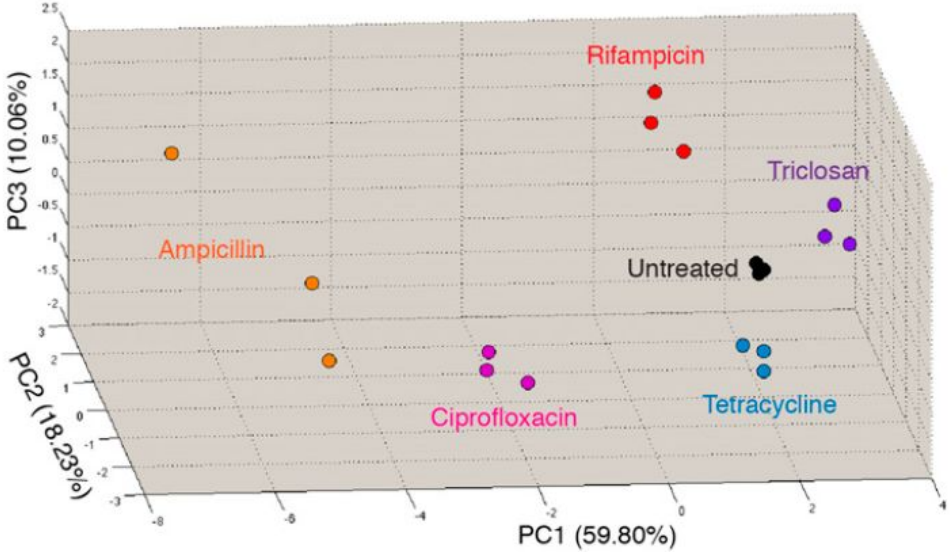


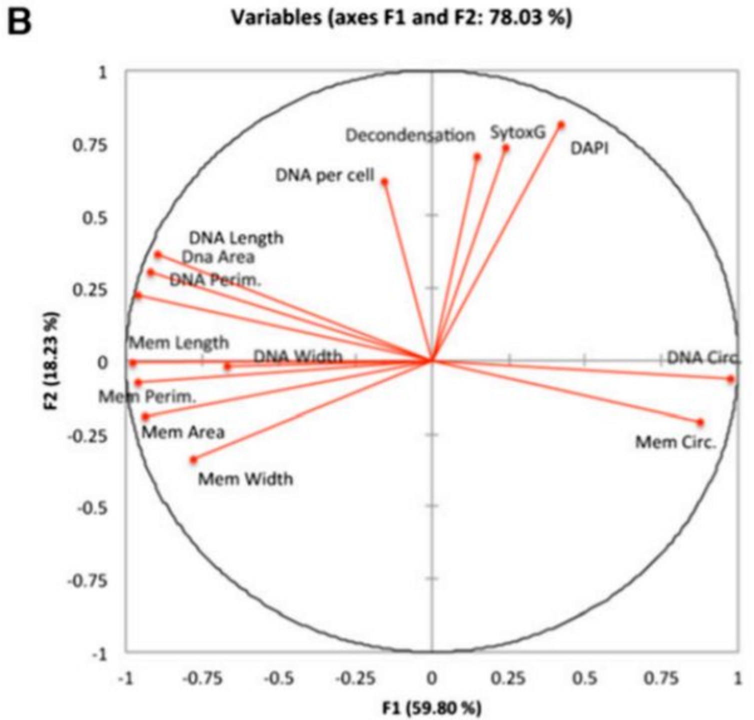
Cytological profiling for antibiotics discovery



P: Protein translation inhibitors
R: RNA transcription inhibitors
D: DNA replication inhibitors
L: Lipid biosynthesis inhibitors
C: Cell-wall synthesis inhibitors (peptidoglycan)

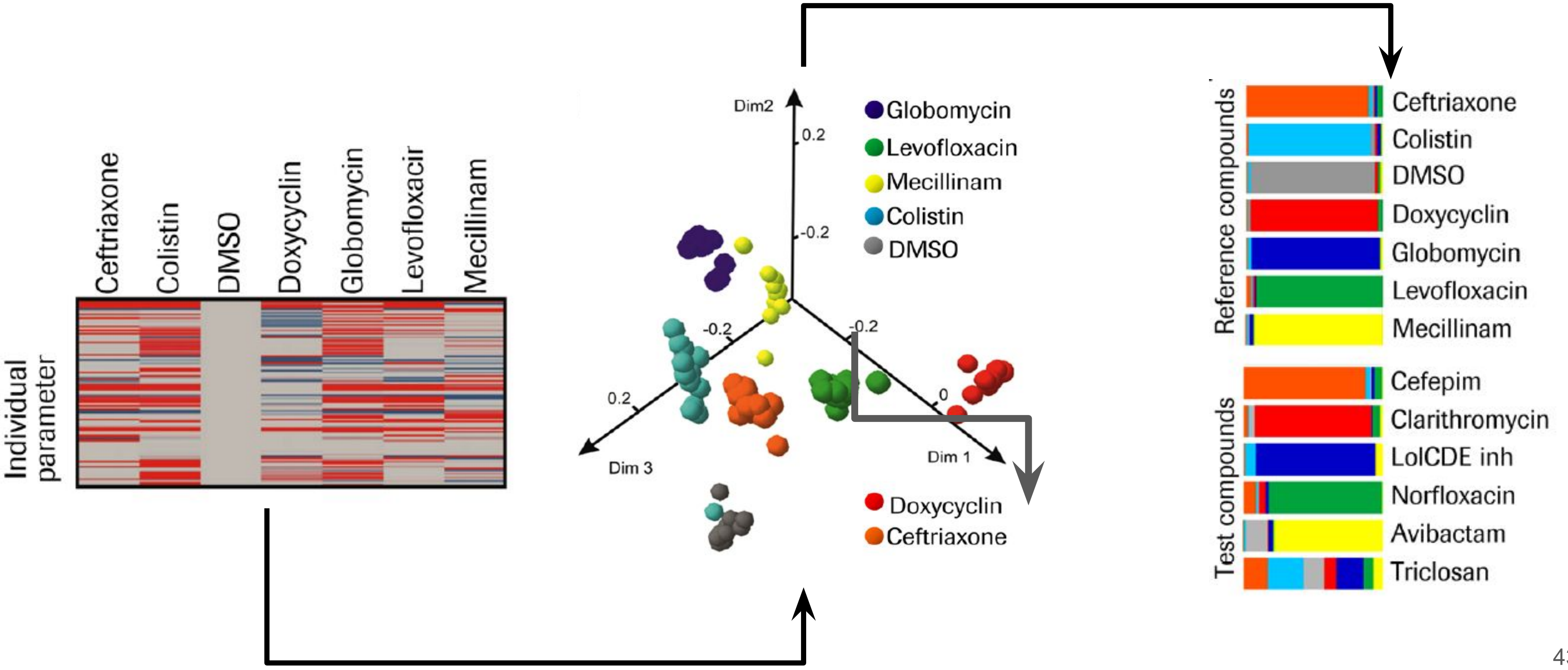
Principal components are linear combination of morphological features



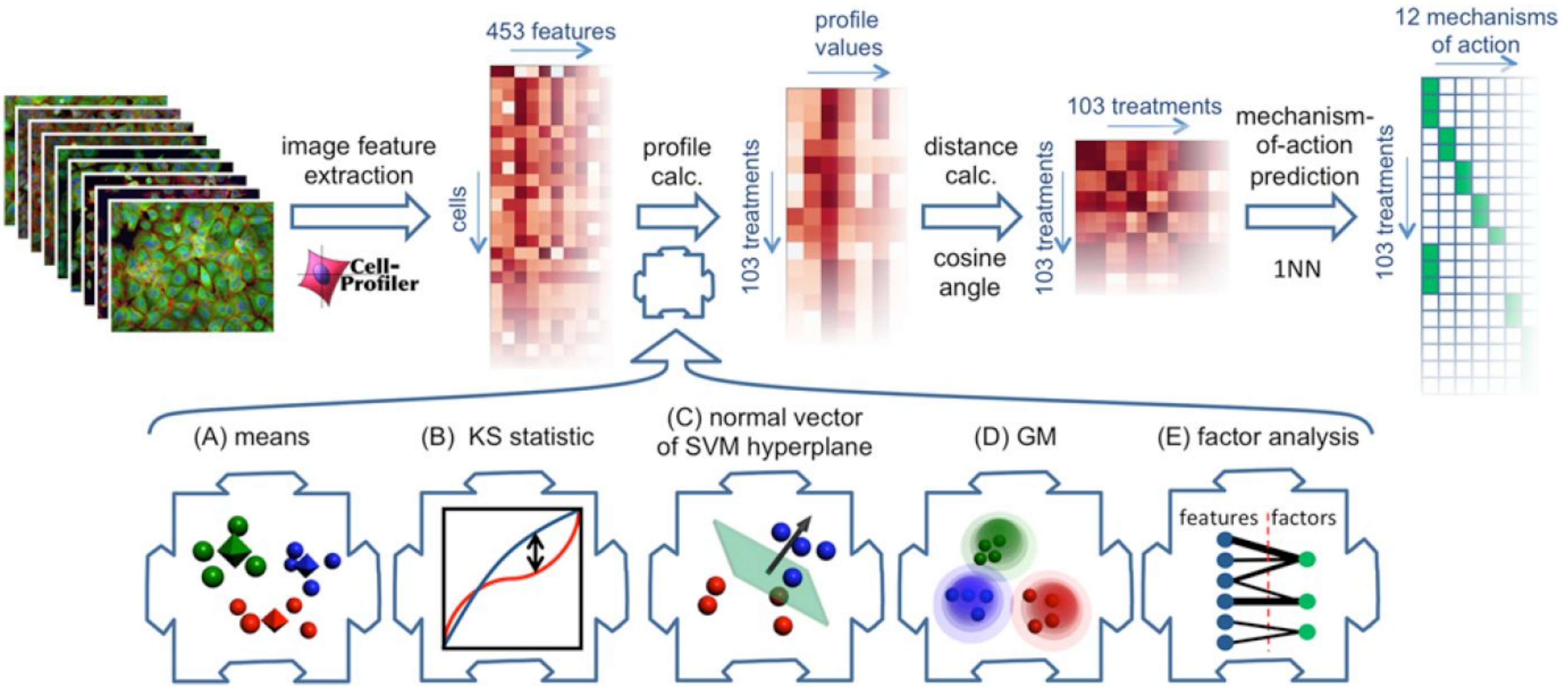


Membrane area, μm^2	DNA area, μm^2	Membrane perimeter, µm	DNA perimeter, µm	Membrane length, µm	DNA length, µm	No. of nucleoids per cell
Membrane width, µm	DNA width, µm	Membrane circularity	DNA circularity	SytoxG intensity	DAPI intensity	Decondensation

Morphology classifies compounds by MoA



Comparison of computational methods



Do the benchmark and use Occam's Razor

Table 1. Accuracies for classifying compound treatments into mechanisms of action.

Method	Accuracy, %
Means	83
KS statistic	83
Normal vector to support-vector machine hyperplane	81
With recursive feature elimination	64
Distribution over Gaussian mixture components	83
Factor analysis + means	94

True mechanistic class		Act	Aur	Ch	DD	DR	Eg5	Epi	KI	MD	MS	PD	PS	Acc.
Actin disruptors	Act	4											1	80 %
Aurora kinase inhibitors	Aur		12											100 %
Cholesterol-lowering	Ch			6										100 %
DNA damage	DD				8									93 %
DNA replication	DR				1	7								81 %
Eg5 inhibitors	Eg5						12							100 %
Epithelial	Epi			2				6						76 %
Kinase inhibitors	KI								5					100 %
Microtubule destabilizers	MD				1					13				93 %
Microtubule stabilizers	MS										9			97 %
Protein degradation	PD											7		96 %
Protein synthesis	PS												8	100 %

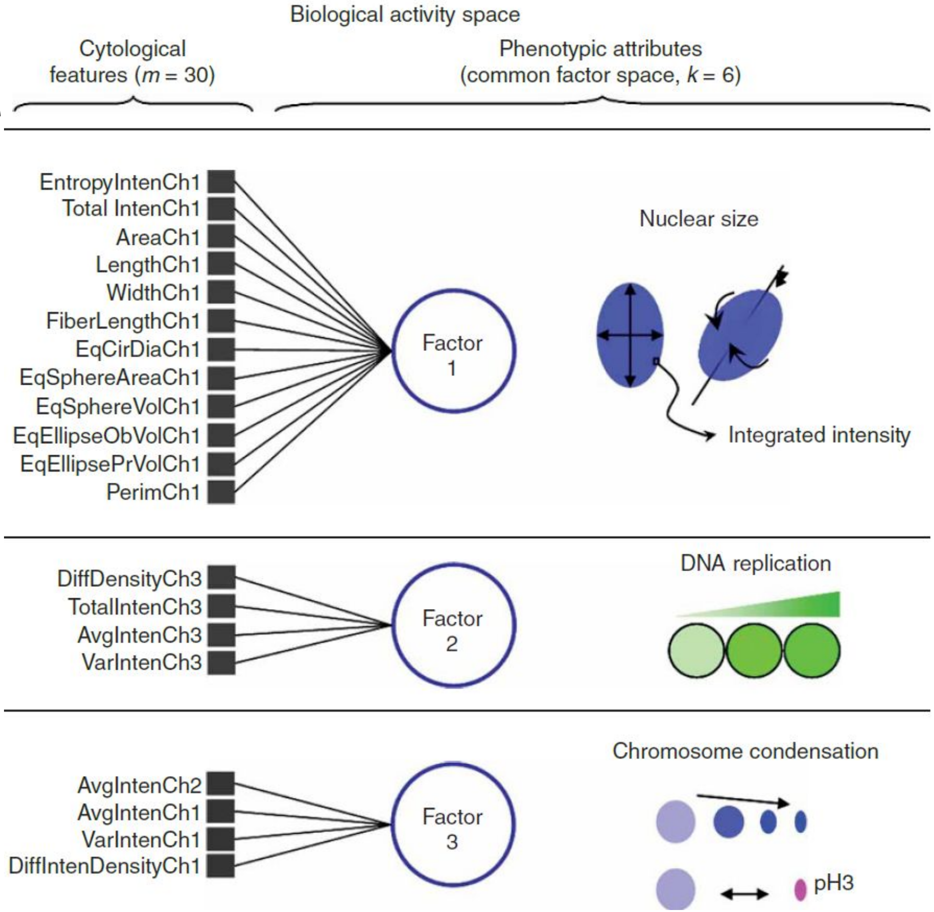
Predicted class (columns). Overall accuracy: **94 %**

A possible explanation for the success of latent variable models

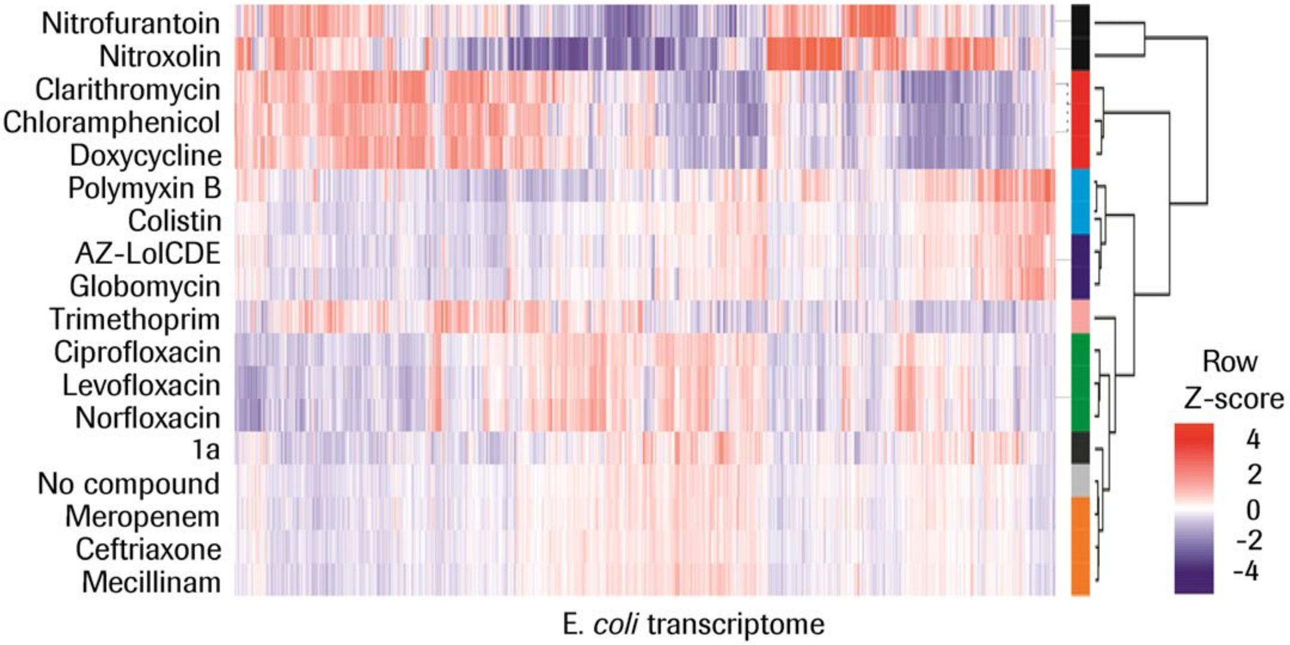
$$
\text{Cells}\begin{pmatrix} x_{11} & \cdots & x_{1m} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{nm} \end{pmatrix} = X_{nm} = \underbrace{\sum_{i=1}^{k}}_{k\text{-factor space}} L_{ni} F_{im} + \varepsilon_{nm}
$$

Cytological features

A common latent factor model



Morphology and gene expression used jointly

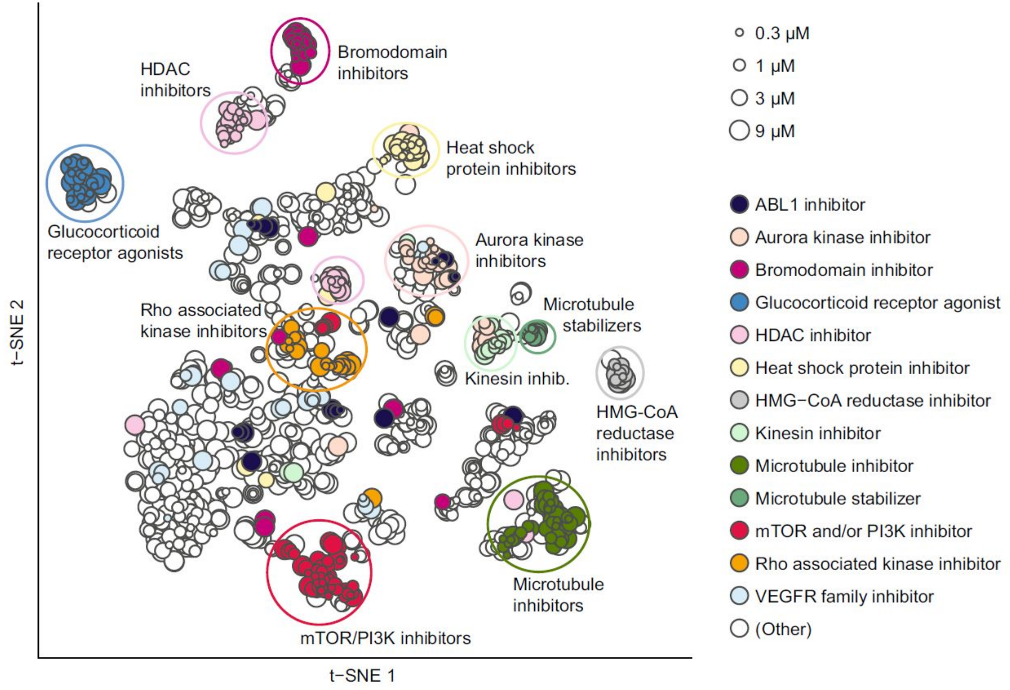


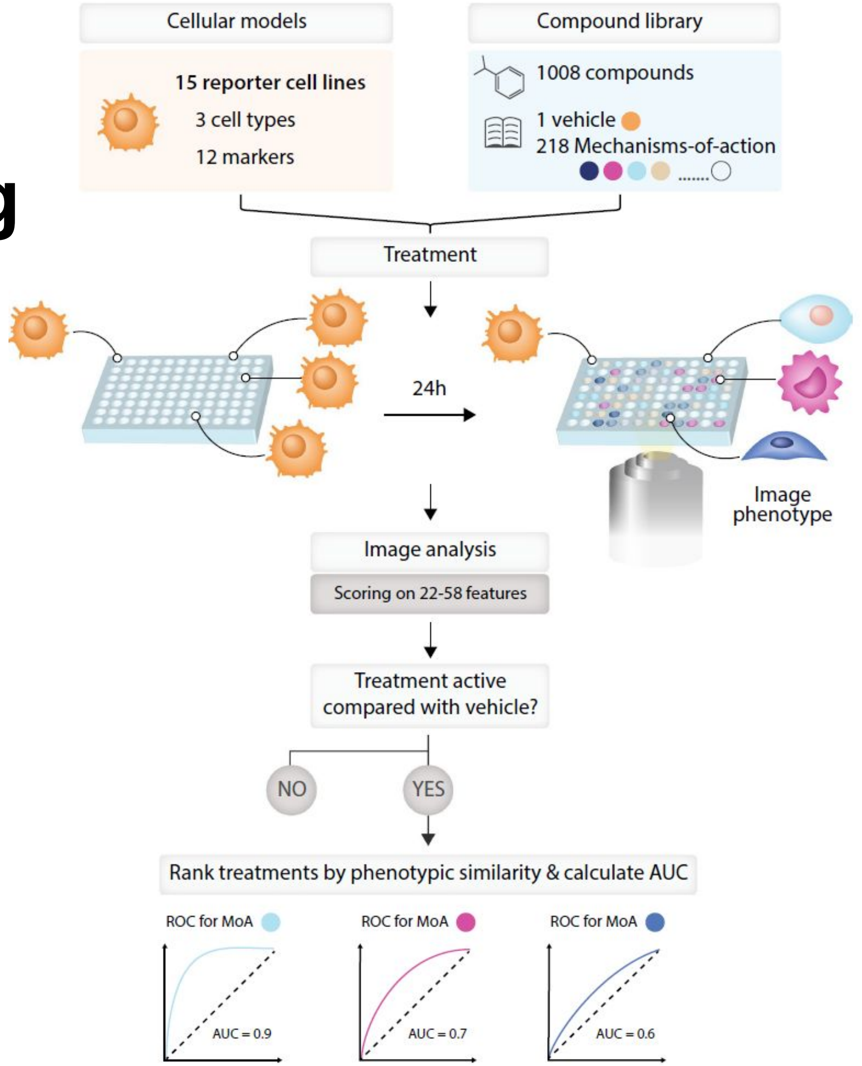
Gene-set enrichment analysis

Reporter assays

Pathway-Phenotype associations

A multi-cell-type, 1008-compound screening by Cox *et al.* (2020)





Conclusions

- Gene expression and image-based profiling can be used individually or jointly for phenotypic screening;
- Integration of biological knowledge, high-throughput data, and statistical modelling empowers phenotypic drug discovery.

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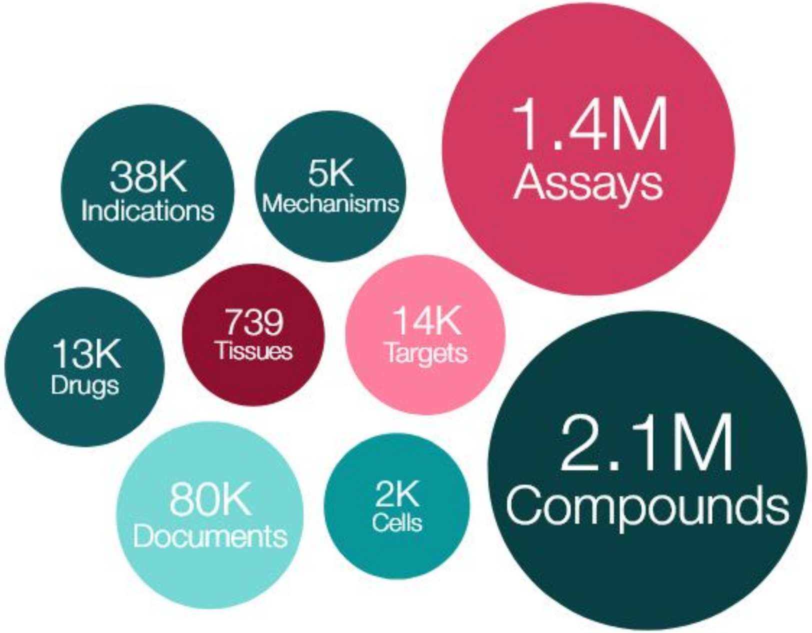
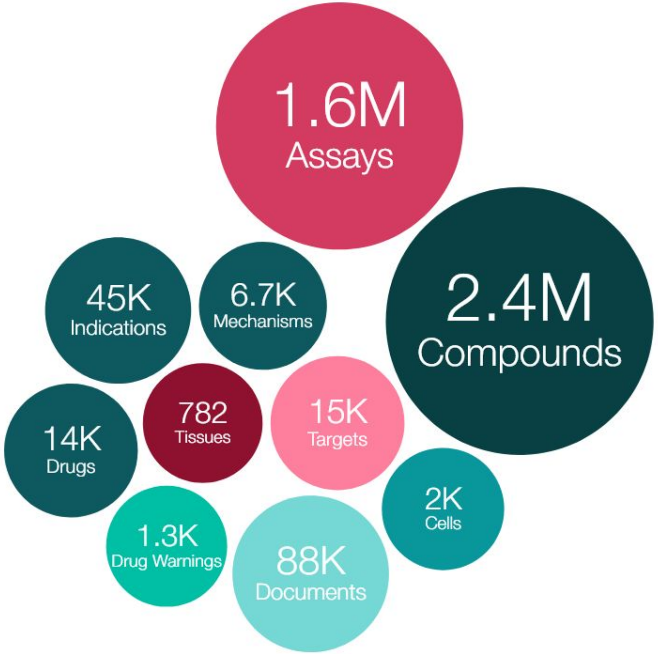
The evolution of ChEMBL database



Visualization of ChEMBL (2021)



[Visualization of ChEMBL](#)
(version 33; 2024)